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Closed-loop identification revisited[1]

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A comprehensive study of closed-loop identification in the prediction error framework is presented. The statistical properties of a number of methods are analysed and it is shown that most of the common methods correspond to different parameterizations of the models.

Abstract

Identification of systems operating in closed loop has long been of prime interest in industrial applications. The problem offers many possibilities, and also some fallacies, and a wide variety of approaches have been suggested, many quite recently. The purpose of the current contribution is to place most of these approaches in a coherent framework, thereby showing their connections and display similarities and differences in the asymptotic properties of the resulting estimates. The common framework is created by the basic prediction error method, and it is shown that most of the common methods correspond to different parameterizations of the dynamics and noise models. The so-called *indirect methods*, e.g., are indeed "direct" methods employing noise models that contain the regulator. The asymptotic properties of the estimates then follow from the general theory and take different forms as they are translated to the particular parameterizations. We also study a new projection approach to closed-loop identification with the advantage of allowing approximation of the open-loop dynamics in a given, and user-chosen frequency domain norm, even in the case of an unknown, nonlinear regulator. © 1999 Elsevier Science Ltd. All rights reserved.

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1. Introduction

1.1. Background and scope

System identification is a well-established field with a number of approaches, that can broadly be classified into the *prediction error family* (e.g., Ljung, 1987), the *subspace approaches* (e.g., Van Overschee and De Moor, 1996), and the nonparametric *correlation and spectral analysis methods* (e.g., Brillinger, 1981). Of special interest is the situation when the data to be used has been collected under closed-loop operation, as in Fig. 1.

The fundamental problem with closed-loop data is the correlation between the unmeasurable noise and the input. It is clear that whenever the feedback controller is not identically zero, the input and the noise will be correlated. This is the reason why several methods that work in open loop fail when applied to closed-loop data. This is, for example, true for the subspace approach and the nonparametric methods, unless special measures are taken. Despite these problems, performing identification experiments under output feedback (i.e., in closed loop) may be necessary due to safety or economic reasons, or if the system contains inherent feedback mechanisms. Closed-loop experiments may also be advantageous in certain situations:

- In Gevers and Ljung (1986) the problem of optimal experiment design is studied. It is shown that if the model is to be used for minimum variance control design the identification experiment should be performed in closed loop with the optimal minimum variance controller in the loop. In general, it can be seen that optimal experiment design with criteria penalizing both the error in the dynamics model and in the noise

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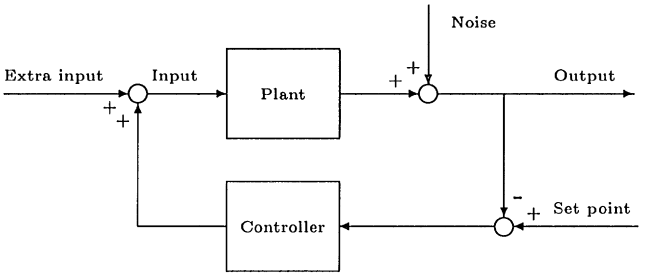


Fig. 1. A closed-loop system

model leads to closed-loop solutions (Ljung, 1987; Hjalmarsson et al., 1996). The same is true with variance constraints on the output (Forssell and Ljung, 1998a).

- In "identification for control" (see, e.g., Zang et al., 1995; Lee et al., 1995; de Callafon et al., 1993) the objective is to achieve a model that is suited for robust control design. Thus, one has to tailor the experiment and preprocessing of data so that the model is reliable in regions where the design process does not tolerate significant uncertainties. The use of closed-loop experiments has been a prominent feature in these approaches.

Historically, there has been a substantial interest in both special identification techniques for closed-loop data, and for analysis of existing methods when applied to such data. One of the earliest results was given by Akaike (1967), who analysed the effect of feedback loops in the system on correlation and spectral analysis. In the 1970s there was a very active interest in questions concerning closed-loop identification, as summarized in the survey paper (Gustavsson et al., 1977). See also Anderson and Gevers (1982). Up to this point much of the attention had been directed towards identifiability and accuracy problems. With the increasing interest "identification for control", the focus has shifted to the ability to shape the bias distribution so that control-relevant model approximations of the system are obtained. The surveys by Gevers (1993) and Van den Hof and Schrama (1995) cover most of the results along this line of research.

It is the purpose of the present paper to "revisit" the area of closed-loop identification, to put some of the new results and methods into perspective, and to give a status report of what can be done and what cannot. In the course of this expose, new results will also be generated.

We will exclusively deal with methods derived in the prediction error framework and most of the results will be given for the multi-input multi-output (MIMO) case. The leading idea in the paper will be to provide a unified framework for many closed-loop methods by treating them as different parameterizations of *the* prediction error method. This will be a useful starting point for a comparative study of the different methods and it allows a simultaneous analysis of the methods using the basic statements on the asymptotic statistical properties of the general prediction error method:

- Convergence and bias distribution of the limit transfer function estimates.
- Asymptotic variance of the transfer function estimates (as the model orders tend to infinity).
- Asymptotic variance and distribution of the parameter estimates.

(For completeness and ease of reference we have included precise formulations of these theoretical results in Appendix A.)

1.2. Approaches to closed-loop identification

In the literature, several different types of closed-loop identification methods have been suggested. We will use the set-up depicted in Fig. 1 to explain the ideas. One may distinguish between methods that

(a) Assume no knowledge about the nature of the feedback mechanism and use only measurements of the plant input, ignoring extra inputs or set-point signals even if known.
(b) Assume the feedback to be known and use possible extra inputs or set-point signals.
(c) Assume the regulator to be unknown, but of a certain structure (like in Fig. 1).

In (b) and (c) it is common to assume that the controller is linear, but the same ideas can also be applied if nonlinear and/or time-varying controllers are used. The price is, of course, that the estimation problems then become much more involved. For simplicity, we will in general assume linear, time-invariant feedback when discussing methods that fall under (b) and (c) in this paper. If the plant input is measured it can be noted that knowing the controller and extra inputs or set-point signals brings no additional information about the open-loop system. Conversely, with a known linear, time-invariant, and noise-free controller, we can easily reconstruct the plant input using measurements of the output and extra inputs or set-point signals and then these carry no further information about the open-loop system. Thus there is no major difference between (a)–(c) when the controller is linear, time-invariant, and noise-free. The problem in industrial practice is rather that no controller has this simple form: various delimiters, anti-windup functions and other nonlinearities will have the controller deviate from the linear, time-invariant form, even if the regulator parameters (e.g. PID-coefficients) are known. This strongly disfavors the second approach.

In this paper, we will use a classification of the different methods that is similar to the one in Gustavsson et al.

(1977). See also Söderström and Stoica (1989). The basis for the classification is the different kinds of possible assumptions on the feedback listed above. The closed-loop identification methods correspondingly fall into the following main groups (cf. Fig. 1):

(1) *The direct approach:* Ignore the feedback and identify the open-loop system using measurements of the input and the output.
(2) *The indirect approach:* Identify some closed-loop transfer function and determine the open-loop parameters using the knowledge of the (linear) controller.
(3) *The joint input–output approach:* Regard the input and output jointly as the output from a system driven by some extra input or set-point signal and noise. Use some method to determine the open-loop parameters from an estimate of this augmented system.

These categories are basically the same as those in Gustavsson et al. (1977), the only difference is that in the joint input–output approach we allow the augmented system to have a measurable input in addition to the unmeasurable noise. The indirect and joint input–output approaches typically assume linear feedback but can also be applied with nonlinear feedback, although these schemes then tend to be very involved.

In the closed-loop identification literature it has been common to classify the methods primarily based on how the final estimates are computed (e.g. directly or indirectly using multi-step estimation schemes), and then the main groupings have been into "direct" and "indirect" methods. This should not, however, be confused with the classification (1)–(3) which is based on the assumptions made on the feedback.

1.3. Outline

The rest of the paper is organized as follows. Next, in Section 2, the assumptions we make on the data-generating mechanism are formalized. This section also introduces some of the notation that will be used in the paper. Section 3 presents the prediction error method and contains a summary of the asymptotic statistical properties of this method. The three basic approaches to closed-loop identification are discussed in Sections 4 and 5. The direct and indirect approaches are covered in Section 4, while Section 5 deals with the joint input–output approach. The asymptotic properties of the different methods are studied in Sections 6–8. In Section 6 convergence results for the limit estimates are given. Section 7 deals with the asymptotic variance of the transfer function estimates as the model orders tend to infinity. The asymptotic distribution of the parameter estimates is studied in Section 8. The last section, Section 9, contains a summarizing discussion on the main results. Two appendices have also been included in the paper. Appendix A contains the basic results for the statistical properties of the prediction error method, here given in full detail. Appendix B contains the deferred proofs from Sections 6–8.

2. Technical assumptions and notation

The basis of all identification is the data set

$$Z^N = \{u(1), y(1), \ldots, u(N), y(N)\} \tag{1}$$

consisting of measured input and output signals $u(t)$ and $y(t)$, $t = 1, \ldots, N$. We will make the following assumptions regarding how this data set was generated.

Assumption 1. *The true system $\mathcal{S}$ is causal, linear, and time-invariant with p outputs and m inputs and given by*

$$y(t) = G_0(q)u(t) + v(t), \qquad v(t) = H_0(q)e(t), \tag{2}$$

where $\{e(t)\}(p \times 1)$ is a zero-mean white-noise process with covariance matrix Λ_0, and bounded moments of order $4 + \delta$, some $\delta > 0$, and $H_0(q)$ is an inversely stable, monic filter.

Assumption 2. *The input $\{u(t)\}$ is given by*

$$u(t) = k(t, y^t, u^{t-1}, r(t)), \tag{3}$$

where $y^t = [y(1), \ldots, y(t)]$, etc., and where the reference signal $\{r(t)\}$ is a given quasi-stationary signal, independent of $\{v(t)\}$ and k is a given deterministic function such that the closed-loop system (2) *and* (3) *is exponentially stable, which we define as follows: For each t, s; $t \geq s$ there exist random variables $\bar{y}_s(t)$, $\bar{u}_s(t)$, independent of r^s and v^s but not independent of r^t and v^t, such that*

$$\bar{E}\,\|y(t) - \bar{y}_s(t)\|^4 < C\lambda^{t-s} \tag{4}$$

$$\bar{E}\,\|u(t) - \bar{u}_s(t)\|^4 < C\lambda^{t-s} \tag{5}$$

for some $C < \infty$, $\lambda < 1$. In addition, k is such that $G_0(q)u(t)$ is independent of $e(s)$ for $s \geq t$.

In the paper q will denote the time-shift operator (i.e., $q^{-1}u(t) = u(t-1)$). The symbol $\bar{E}$ is defined as

$$\bar{E}f(t) = \lim_{N\to\infty} \frac{1}{N} \sum_{t=1}^{N} Ef(t). \tag{6}$$

Unless otherwise stated, we will assume that all signals are quasi-stationary (Ljung, 1987). This means that we can define spectra and cross-spectra of signals, which will be denoted $\Phi_u(\omega)$, $\Phi_{ue}(\omega)$, etc. We will assume that we can factorize the $\Phi_u(\omega)$ (and all other signal spectra) as

$$\Phi_u(\omega) = M_u(e^{i\omega})M_u^T(e^{-i\omega}) \tag{7}$$

where $M_u(e^{i\omega})$ is causal, stable, and inversely stable.

Introduce the signal $\chi_0(t) = [u^T(t)\ e^T(t)]^T$. The spectrum is

$$\Phi_{\chi_0} = \begin{bmatrix} \Phi_u & \Phi_{ue} \\ \Phi_{eu} & \Lambda_0 \end{bmatrix}. \tag{8}$$

Here we have suppressed the argument ω which also will be done in the sequel whenever there is no risk of confusion. Similarly, we will also frequently suppress the arguments t, q, and $e^{i\omega}$ for notational convenience. The following factorizations of Φ_{χ_0} will be useful (it is assumed that $\Phi_u > 0$ and $\Lambda_0 > 0$):

$$\Phi_{\chi_0} = \begin{bmatrix} I & \Phi_{ue}\Lambda_0^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} \Phi_u^r & 0 \\ 0 & \Lambda_0 \end{bmatrix} \begin{bmatrix} I & 0 \\ \Lambda_0^{-1}\Phi_{eu} & I \end{bmatrix} \tag{9}$$

$$= \begin{bmatrix} I & 0 \\ \Phi_{eu}\Phi_u^{-1} & I \end{bmatrix} \begin{bmatrix} \Phi_u & 0 \\ 0 & \Phi_e^r \end{bmatrix} \begin{bmatrix} I & \Phi_u^{-1}\Phi_{ue} \\ 0 & I \end{bmatrix}. \tag{10}$$

Here

$$\Phi_u^r = \Phi_u - \Phi_{ue}\Lambda_0^{-1}\Phi_{eu}, \tag{11}$$

$$\Phi_e^r = \Lambda_0 - \Phi_{eu}\Phi_u^{-1}\Phi_{ue}. \tag{12}$$

We will also use

$$\Phi_{\chi_0} = \Phi_{\chi_0}^r + \Phi_{\chi_0}^e, \qquad \Phi_{\chi_0}^r = \begin{bmatrix} \Phi_u^r & 0 \\ 0 & 0 \end{bmatrix},$$

$$\Phi_{\chi_0}^e = \begin{bmatrix} \Phi_u^e & \Phi_{ue} \\ \Phi_{eu} & \Lambda_0 \end{bmatrix}, \tag{13}$$

$$\Phi_u^e = \Phi_u - \Phi_u^r = \Phi_{ue}\Lambda_0^{-1}\Phi_{eu}. \tag{14}$$

The matrix Φ_u^r defined in Eq. (11) can be seen as that part of the input spectrum that originates from the reference signal r. Similary, Φ_u^e defined in Eq. (14) can be seen as that part of the input spectrum that originates from the noise e.

For some of the analytic treatment we shall assume that the input u is generated as

$$u(t) = r(t) - K(q)y(t), \tag{15}$$

where K is a linear, time-invariant regulator of appropriate dimensions and where r is the reference signal (e.g., an extra input or a set-point signal), which is assumed independent of the noise v. If Eq. (15) holds then Assumption 2 simply means that the closed-loop system is asymptotically stable.

Let us now introduce some further notation for the linear feedback case. By combining Eqs. (2) and (15) we get the closed-loop relations

$$y(t) = S_0(q)G_0(q)r(t) + S_0(q)v(t), \tag{16}$$

$$u(t) = S_0^i(q)r(t) - S_0^i(q)K(q)v(t), \tag{17}$$

where S_0 and S_0^i are the output and input sensitivity functions, respectively,

$$S_0(q) = (I + G_0(q)K(q))^{-1}, \tag{18}$$

$$S_0^i(q) = (I + K(q)G_0(q))^{-1}. \tag{19}$$

For future use we also introduce

$$G_0^c(q) = S_0(q)G_0(q) \quad \text{and} \quad H_0^c(q) = S_0(q)H_0(q) \tag{20}$$

so that we can rewrite the closed-loop system (16) as

$$y(t) = G_0^c(q)r(t) + v_c(t), \qquad v_c(t) = H_0^c(q)e(t) \tag{21}$$

Using Eq. (17) we can easily derive the following expressions for Φ_u^r and Φ_u^e under Eq. (15):

$$\Phi_u^r = S_0^i\Phi_r(S_0^i)^*, \tag{22}$$

$$\Phi_u^e = S_0^iK\Phi_vK^*(S_0^i)^* = KS_0\Phi_vS_0^*K^*. \tag{23}$$

Here Φ_r is the spectrum of the reference signal and $\Phi_v = H_0\Lambda_0H_0^*$ the noise spectrum. Superscript $*$ denotes complex conjugate transpose. If u is generated as in Eq. (15) the cross-spectrum between u and e is

$$\Phi_{ue} = -KS_0H_0\Lambda_0 = -S_0^iKH_0\Lambda_0. \tag{24}$$

Occasionally, we shall also consider the case where the regulator is linear as in Eq. (15) but contains an unknown additive disturbance d:

$$u(t) = r(t) - K(q)y(t) + d(t). \tag{25}$$

We will assume that the nature of d is such that the closed-loop system (2), Eq. (25) is exponentially stable in the sense defined in Assumption 2. The origin of the disturbance d can, e.g., be an imperfect knowledge of the true regulator parameters: Suppose that the true regulator is given by $K + \Delta_K$ where K is known and Δ_K is unknown. We could then use the model (25) to describe how the input was generated. (In this case $d = -\Delta_K y$.) It is also worth noting that if K is known and both u and y are measured we can regard d as part of the external excitation since then we can always construct

$$r'(t) = u(t) + K(q)y(t) (= r(t) + d(t)) \tag{26}$$

and replace r by this signal in our model. This would lead us back to the situation (15); see, e.g., Van den Hof (1997).

3. Prediction error identification

In this section we shall review some basic results on prediction error methods, that will be used in the sequel. See Appendix A and Ljung (1987) for more details.

3.1. The method

We will work with a model structure $\mathscr{M}$ of the form

$$y(t) = G(q, \theta)u(t) + H(q, \theta)e(t). \tag{27}$$

G will be called the dynamics model and H the noise model. Unless otherwise explicitly stated, we will assume that G is causal and such that $G(q, \theta)u(t)$ is independent of $e(s)$ for $s \geq t$ and that H is monic and causal. The parameter vector θ ranges over a set $D_{\mathscr{M}}$ which is assumed compact and connected. The one-step-ahead predictor for the model structure (27) is (Ljung, 1987)

$$\hat{y}(t|\theta) = H^{-1}(q, \theta)G(q, \theta)u(t) + (I - H^{-1}(q, \theta))y(t). \tag{28}$$

The prediction error is

$$\varepsilon(t, \theta) = y(t) - \hat{y}(t|\theta) = H^{-1}(q, \theta)(y(t) - G(q, \theta)u(t)). \tag{29}$$

Given the model (28) and measured data Z^N we determine the prediction error estimate through

$$\hat{\theta}_N = \arg\min_{\theta \in D_{\mathscr{M}}} V_N(\theta, Z^N), \tag{30}$$

$$V_N(\theta, Z^N) = \frac{1}{N}\sum_{t=1}^{N} \varepsilon_F^{\mathrm{T}}(t, \theta)\Lambda^{-1}\varepsilon_F(t, \theta), \tag{31}$$

$$\varepsilon_F(t, \theta) = L(q, \theta)\varepsilon(t, \theta). \tag{32}$$

Here Λ is a symmetric, positive definite weighting matrix and L a (possibly parameter-dependent) monic prefilter that can be used to enhance certain frequency regions. It is easy to see that

$$\varepsilon_F(t, \theta) = L(q, \theta)H^{-1}(q, \theta)(y(t) - G(q, \theta)u(t)). \tag{33}$$

Thus the effect of the prefilter L can be included in the noise model and $L(q, \theta) = I$ can be assumed without loss of generality. This will be done in the sequel. The resulting transfer function (matrix) estimates will be denoted $\hat{G}_N(q)$, $\hat{G}_N(q) = G(q, \hat{\theta}_N)$, etc.

We say that the true system is contained in the model set if, for some $\theta_0 \in D_{\mathscr{M}}$,

$$G(q, \theta_0) = G_0(q), \qquad H(q, \theta_0) = H_0(q). \tag{34}$$

This will also be written $\mathscr{S} \in \mathscr{M}$. The case when the true noise properties cannot be correctly described within the model set but where there exists a $\theta_0 \in D_{\mathscr{M}}$ such that

$$G(q, \theta_0) = G_0(q) \tag{35}$$

will be denoted $G_0 \in \mathscr{G}$.

3.2. Convergence

Define the average criterion $\bar{V}(\theta)$ as

$$\bar{V}(\theta) = \bar{E}\varepsilon^{\mathrm{T}}(t, \theta)\Lambda^{-1}\varepsilon(t, \theta). \tag{36}$$

Then we have the following result (see, e.g., Ljung, 1978, 1987):

$$\hat{\theta}_N \to D_{\mathrm{c}} = \arg\min_{\theta \in D_{\mathscr{M}}} \bar{V}(\theta) \quad \text{with probability (w.p.) 1 as } N \to \infty. \tag{37}$$

Note that we can write $\bar{V}(\theta)$ as

$$\bar{V}(\theta) = \bar{E}\,\mathrm{tr}[\Lambda^{-1}\varepsilon(t, \theta)\varepsilon^{\mathrm{T}}(t, \theta)] = \frac{1}{2\pi}\int_{-\pi}^{\pi} \mathrm{tr}[\Lambda^{-1}\Phi_\varepsilon]\,\mathrm{d}\omega. \tag{38}$$

Here Φ_ε is the spectrum of the prediction error $\varepsilon(t, \theta)$. It can thus be seen that D_{c} defined in Eq. (37) depends only on the second-order properties of (the spectrum) of the prediction error.

In case the input–output data can be described by Eq. (2) we have the following characterization of D_{c} (G_θ is short for $G(q, \theta)$, etc.):

$$D_{\mathrm{c}} = \arg\min_{\theta \in D_{\mathscr{M}}} \int_{-\pi}^{\pi} \mathrm{tr}\left[[(G_0 - G_\theta) \quad (H_0 - H_\theta)]\Phi_{\chi_0} \times \begin{bmatrix} (G_0 - G_\theta)^* \\ (H_0 - H_\theta)^* \end{bmatrix} (H_\theta \Lambda H_\theta^*)^{-1} \right] \mathrm{d}\omega \tag{39}$$

where Φ_{χ_0} is given by Eq. (8). This is shown in Appendix (A.1). Note that the result holds regardless of the nature of the regulator, as long as Assumptions 1 and 2 hold and the signals involved are quasistationary (which is a standing assumption).

From Eq. (39) several conclusions regarding the consistency of the method can be drawn. First of all, suppose that the parameterization of G and H is sufficiently flexible so that $\mathscr{S} \in \mathscr{M}$. If this holds, the method will give consistent estimates of G_0 and H_0 if the experiment is *informative* Ljung (1987), which means that the matrix Φ_{χ_0} is positive definite for all frequencies. (Note that it will always be positive *semi*-definite since it is a spectral matrix.) From the factorization (9) we see that this holds if, and only if, Φ_u^r defined in Eq. (11) is positive definite. The general condition for this to be true is that there should *not* be a linear, time-invariant, and noise-free relationship between u and y. With an external reference signal this is automatically satisfied, but it should be clear that informative closed-loop experiments can also be guaranteed if we switch between different linear regulators or use a nonlinear regulator (see, e.g., Gustavsson et al., 1977; Ljung, 1987). With a linear controller we need a persistently exciting reference signal to ensure that Φ_u^r is positive definite. To see this note that, in the linear case, $\Phi_u^r = S_0^{\mathrm{i}}\Phi_r(S_0^{\mathrm{i}})^*$ so Φ_u^r is positive definite for all frequencies if, and only if, Φ_r is (which is the same as to say that the reference signal is persistently exciting (Ljung, 1987)), since the analytical function S_0^{i} (cf. Eq. (22)) can be zero at most finitely many points.

It can also be noted that the prediction error method can give consistent estimates even if the experiment is not informative, depending on the parameterization of G and H: If $\mathscr{S} \in \mathscr{M}$ the method still gives consistent estimates of G_0 and H_0 even though Φ_{χ_0} is not positive definite provided that $[(G_0 - G_\theta)\ (H_0 - H_\theta)]$ does not lie in the (left) null space of Φ_{χ_0}. In Ljung (1987) this is called that the experiment is *informative enough* with respect to the chosen model structure.

3.3. Asymptotic variance of black box transfer function estimates

Consider the model structure (27). Introduce

$$T(q, \theta) = \mathrm{vec}[G(q, \theta) \quad H(q, \theta)]. \tag{40}$$

(The vec-operator stacks the columns of its argument on top of each other in a vector. A more formal definition is given in Appendix A.2.) Suppose that the vector θ can be decomposed so that

$$\theta = [\theta_1^{\mathrm{T}}, \theta_2^{\mathrm{T}}, \ldots, \theta_n^{\mathrm{T}}]^{\mathrm{T}} \quad \dim \theta_k = s, \quad \dim \theta = n \cdot s. \tag{41}$$

We shall call n the order of the model structure (27) and for some results we will allow n to tend to infinity as N tends to infinity. Suppose also that T in Eq. (40) has the following shift structure:

$$\frac{\partial}{\partial \theta_k} T(q, \theta) = q^{-k+1} \frac{\partial}{\partial \theta_1} T(q, \theta). \tag{42}$$

It should be noted that most polynomial-type model structures, including the ones studied in this paper, satisfy this shift structure. Thus, Eq. (42) is a rather weak assumption.

More background material including further technical assumptions and additional notation can be found in Appendix A.2. For brevity reasons, we here go directly to the main result ($\otimes$ denotes the Kronecker product):

$$\mathrm{Cov}\,\mathrm{vec}[\hat{T}_N(\mathrm{e}^{\mathrm{i}\omega})] \sim \frac{n}{N} (\Phi_{\chi_0}(\omega))^{-\mathrm{T}} \otimes \Phi_v(\omega). \tag{43}$$

The covariance matrix is thus proportional to the model order n divided by the number of data N. This holds asymptotically as both n and N tend to infinity. In open loop we have $\Phi_{ue} = 0$ and

$$\mathrm{Cov}\,\mathrm{vec}[\hat{G}_N(\mathrm{e}^{\mathrm{i}\omega})] \sim \frac{n}{N} (\Phi_u(\omega))^{-\mathrm{T}} \otimes \Phi_v(\omega), \tag{44}$$

$$\mathrm{Cov}\,\mathrm{vec}[\hat{H}_N(\mathrm{e}^{\mathrm{i}\omega})] \sim \frac{n}{N} \Lambda_0^{-1} \otimes \Phi_v(\omega). \tag{45}$$

Notice that the result (44) for the dynamics model holds also in case the noise model is fixed (e.g., $H(q, \theta) = I$).

3.4. Asymptotic distribution of the parameter vector estimates

If $\mathscr{S} \in \mathscr{M}$ then $\hat{\theta}_N \to \theta_0$ as $N \to \infty$ under reasonable conditions on the data set (Ljung, 1987). Then, if $\Lambda = \Lambda_0$,

$$\sqrt{N}(\hat{\theta}_N - \theta_0) \in AsN(0, P_\theta), \tag{46a}$$

$$P_\theta = [\bar{E}\psi(t, \theta_0)\Lambda_0^{-1}\psi^{\mathrm{T}}(t, \theta_0)]^{-1}, \tag{46b}$$

where ψ is the negative gradient of the prediction error ε with respect to θ.

In this paper we will restrict to the SISO case when discussing the asymptotic distribution of the parameter vector estimates for notational convenience. For ease of reference we have, in Appendix A.3, stated a variant of Eq. (46) as a theorem.

4. The direct and indirect approaches

4.1. The direct approach

The direct approach amounts to applying a prediction error method directly to input–output data, ignoring possible feedback. This implies that this method can be applied to system with arbitrary (unknown) feedback mechanisms. In general, one works with model structures of the form (27), that is

$$y(t) = G(q, \theta)u(t) + H(q, \theta)e(t) \tag{47}$$

The direct method can thus be formulated as in Eqs. (30)–(32) and coincides with the standard (open-loop) prediction error method (e.g., Ljung, 1987; Söderström and Stoica, 1989).

It is sometimes claimed that the direct approach cannot be used if the underlying system is unstable. This is not correct: The prediction error method can be applied directly also to unstable systems as long as the *predictor* is stable. This is automatically satisfied with ARX and ARMAX models and can also be guaranteed for the output error and Box–Jenkins model structures if special measures are taken (Forssell and Ljung, 1998b). Let us now turn to the indirect approach.

4.2. The idea behind the indirect approach

If the controller and some extra input or set-point signal (cf. Fig. 1) are known we may use indirect identification. This method can be applied to systems with nonlinear feedback, but to explain the idea it is easier to restrict to linear feedback. This has also been a standard assumption in the literature. Suppose therefore that the input is generated as in Eq. (15). The closed-loop system is (cf. Eq. (21)):

$$y(t) = G_0^{\mathrm{c}}(q)r(t) + v_{\mathrm{c}}(t), \tag{48}$$

where $G_0^c = (I + G_0K)^{-1}G_0$. If the regulator K is known and r is measurable, we can identify the closed-loop system, e.g., using the prediction error method with a model

$$y(t) = G^c(q, \theta)r(t) + H_*(q)e(t) \tag{49}$$

and compute an estimate $\hat{G}_N$ of the open-loop system by solving the equation

$$\hat{G}_N^c(q) = (I + \hat{G}_N(q)K(q))^{-1}\hat{G}_N(q). \tag{50}$$

This is the basic idea behind the indirect method and illustrates the some of the features of this method. For instance, note that by identifying the closed-loop system we have transformed the closed-loop identification problem into an "open loop" one, since *r and v* are uncorrelated. This means that we may use any method that works in open loop to find a model of the closed-loop system, for instance a subspace method or instrumental variables. Here we will focus on prediction error methods.

We have outlined a procedure consisting of two steps: First the closed-loop system is identified, then an open-loop model is computed. It should be realized, however, that there are alternative ways of implementing this method. For instance, instead of identifying the closed-loop system we may also identify the sensitivity function using measurements of u and r (see de Callafon and Van den Hof, 1996). Here we will limit the discussion to methods that aim at identifying the closed-loop system. The prediction error approach allows arbitrary parameterizations of the models, so if this method is used it is also possible to construct different indirect "methods" that suit our purposes by considering different parameterizations. Such ideas will be studied next.

4.3. Indirect identification using the prediction error method

With the indirect method we can use output error models with fixed noise models/prefilters to shape the bias distribution (cf. Corollary 6 below). The model structure would then be given by Eq. (49) where $G^c(q, \theta)$ is a parameterized model of the closed-loop system. This gives us an estimate $\hat{G}_N^c$ of the closed-loop system and the open-loop estimate can then be found using Eq. (50). The exact solution is

$$\hat{G}_N(q) = \hat{G}_N^c(q)(I - \hat{G}_N^c(q)K(q))^{-1}. \tag{51}$$

This is of course a perfectly valid solution, although an estimate $\hat{G}_N$ computed this way will generally be of high order. (The order will typically be equal to the sum of the orders of $\hat{G}_N^c$ and K.) If we attempt to solve Eq. (50) with the additional constraint that $\hat{G}_N$ should be of a certain (low) order, we end up with an over-determined system of equations which can be solved in many ways, for instance in a weighted least-squares sense. Examples of such indirect "methods" will be given in Section 8.2 below. A particularly simple and attractive approach is to let the parameters θ relate to properties of the open-loop system G, so that in the first step we should parameterize $G^c(q, \theta)$ as

$$G^c(q, \theta) = (I + G(q, \theta)K(q))^{-1}G(q, \theta). \tag{52}$$

(This was apparently first suggested as an exercise in Ljung (1987). See also van Donkelaar and Van den Hof (1997).) The advantage with this parameterization is that the (problematic) second step (50) is avoided and an open-loop model $\hat{G}_N$ is delivered directly.

The parameterization of $G^c(q, \theta)$ may thus be important for numeric and algebraic reasons, but we emphasize that it does not affect the asymptotic statistical properties of the estimate $\hat{G}_N^c(q)$, as long as it covers the same set of transfer functions.

Another nice and interesting parameterization idea is to use the, so called, dual Youla parameterization that parameterizes all systems that are stabilized by a certain regulator K (see, e.g., Vidyasagar, 1985). This will be studied next.

4.3.1. The dual Youla parameterization

To present the idea, the concept of coprime factorizations of transfer functions is required: a pair of stable transfer functions N, D is a right coprime factorization (rcf) of G if $G = ND^{-1}$ and there exist stable transfer functions X, Y such that $XN + YD = I$. The dual Youla parameterization now works as follows. Let G_{nom} with rcf(N, D) be any system that is stabilized by K with rcf(X, Y). Then, as R ranges over all stable transfer functions, the set

$$\{G : G(q, \theta) = (N(q) + Y(q)R(q, \theta))(D(q) - X(q)R(q, \theta))^{-1}\} \tag{53}$$

describes all systems that are stabilized by K. The unique value of R that corresponds to the true plant G_0 is given by

$$R_0(q) = Y^{-1}(q)(I + G_0(q)K(q))^{-1}(G_0(q) - G_{\text{nom}}(q))D(q). \tag{54}$$

This idea can now be used for identification (see, e.g., Hansen,1989; Hansen et al., 1989; de Callafon and Van den Hof, 1996): Given an estimate $\hat{R}_N$ of R_0 we can compute an estimate of G_0 as

$$\hat{G}_N(q) = (N(q) + Y(q)\hat{R}_N(q))(D(q) - X(q)\hat{R}_N(q))^{-1}. \tag{55}$$

Using the dual Youla parameterization we can write

$$G^c(q, \theta) = (N(q) + Y(q)R(q, \theta))(D(q) + X(q)Y^{-1}(q)N(q))^{-1} \tag{56}$$

$$\triangleq (N(q) + Y(q)R(q, \theta))M(q). \tag{57}$$

With this parameterization the identification problem (49) becomes

$$z(t) = R(q, \theta)x(t) + \bar{v}_c(t), \tag{58}$$

where

$$z(t) = Y^{-1}(q)(y(t) - N(q)M(q)r(t)), \tag{59}$$

$$x(t) = M(q)r(t), \tag{60}$$

$$\bar{v}_c(t) = Y^{-1}(q)v_c(t). \tag{61}$$

Thus, it can be seen that the dual Youla method is a special parameterization of the general indirect method. This means, as mentioned, that the statistical properties of the resulting estimates for the indirect method remain unaffected for the dual Youla method. The main advantage of this method is of course that the obtained estimate $\hat{G}_N$ is guaranteed to be stabilized by K, which clearly is a nice feature. A drawback is that this method typically will give high-order estimates — typically the order will be equal to the sum of the orders of G_{nom} and $\hat{R}_N$.

Before turning to the joint input–output approach, let us pause and study an interesting variant of the parameterization idea used in Eq. (52) which will provide useful insights into the connection between the direct and indirect methods.

4.4. A formal connection between direct and indirect methods

The noise model H in a linear dynamics model structure has often turned out to be a key to interpretation of different "methods". The distinction between the models/"methods" ARX, ARMAX, output error, Box–Jenkins, etc., is entirely explained by the choice of the noise model. Also the practically important feature of prefiltering is equivalent to changing the noise model. Even the choice between minimizing one- or k-step prediction errors can be seen as a noise model issue. See, e.g., Ljung (1987) for all this.

Therefore it should not come as a surprise that also the distinction between the fundamental approaches of direct and indirect identification can be seen as a choice of noise model.

The idea is to use a model parameterized as

$$y(t) = G(q, \theta)u(t) + (I + G(q, \theta)K(q))H_1(q, \theta)e(t). \tag{62}$$

We thus link the noise model to the dynamics model. There is nothing strange with that: so do ARX and ARMAX models. The parameterization (62) is thus perfectly valid, but it should still be pointed out that this is a highly specialized parameterization using the knowledge of K. Also note that this particular parameterization scales H_1 with the inverse model sensitivity function. Similar parameterization ideas have been studied in, e.g., Åström (1993), Landau and Boumaiza (1996), Egardt (1997) and van Donkelaar and Van den Hof (1997).

The predictor for Eq. (62) is

$$\hat{y}(t\,|\,\theta) = H^{-1}(q, \theta)G(q, \theta)u(t) + (I - H^{-1}(q, \theta))y(t). \tag{63}$$

Using $u = r - Ky$ and inserting Eq. (62) we get

$$\hat{y}(t\,|\,\theta) = H_1^{-1}(q, \theta)(I + G(q, \theta)K(q))^{-1}G(q, \theta)r(t) + (I - H_1^{-1}(q, \theta))y(t). \tag{64}$$

But this is exactly the predictor also for the closed-loop model structure

$$y(t) = (I + G(q, \theta)K(q))^{-1}G(q, \theta)r(t) + H_1(q, \theta)e(t) \tag{65}$$

and hence the two approaches are equivalent. We formulate this result as a lemma:

Lemma 3. *Suppose that the input is generated as in Eq.* (15) *and that both u and r are measurable and that the linear regulator K is known. Then, applying a prediction error method to Eq.* (62) *or to Eq.* (65) *gives identical estimates* $\hat{\theta}_N$. *This holds regardless of the parameterization of G and* H_1.

Among other things, this shows that we can use any theory developed for the direct approach (allowing for feedback) to evaluate properties of the indirect approach, and vice versa. It can also be noted that the particular noise model used in Eq. (62) is the answer to the question how H should be parameterized in the direct method in order to avoid the bias in the G-estimate in the case of closed-loop data, even if the true noise characteristics is not correctly modeled. This is further studied in Ljung and Forssell (1999, to appear).

After having discussed so many variants of the indirect method it is perhaps in place to remark that we will refer to the model structure (49) with G^c is parameterized as in Eq. (52) when discussing the indirect method in the sequel. The only exception will be Section (8.2) where we discuss a different implementation of the indirect method.

5. The joint input–output approach

The third main approach to closed-loop identification is the so-called joint input–output approach. The basic assumption in this approach is that the input is generated using a regulator of a certain form, e.g., Eq. (15). Exact knowledge of the regulator parameters is not required — an advantage over the indirect method where this is a necessity.

5.1. The idea

Suppose that the feedback is linear and given by Eq. (15). Closed-loop expressions for the output and the input were given in Eqs. (16) and (17), respectively,

$$y(t) = G_0^c(q)r(t) + S_0(q)v(t), \tag{66}$$

$$u(t) = S_0^i(q)r(t) - S_0^i(q)K(q)v(t). \tag{67}$$

The basic idea in the joint input–output approach is to compute estimates of the open-loop system using estimates of the different transfer functions in Eqs. (66) and (67). A straightforward approach would be to identify the closed-loop system and the input sensitivity function and form the estimate $\hat{G}_N = \hat{G}_N^c(\hat{S}_N^i)^{-1}$. (The rationale behind this is the relation $G_0^c(S_0^i)^{-1} = G_0$.) This idea really goes back to Akaike (1967) who showed that spectral analysis of closed-loop data should be performed as follows: compute the spectral estimates (SISO)

$$\hat{G}_N^{yx} = \frac{\hat{\Phi}_{yx}^N}{\hat{\Phi}_x^N} \quad \text{and} \quad \hat{G}_N^{ux} = \frac{\hat{\Phi}_{ux}^N}{\hat{\Phi}_x^N}, \tag{68}$$

where the signal x is correlated with y and u, but uncorrelated with the noise e (a standard choice is $x = r$) and estimate the open-loop system as

$$\hat{G}_N^{yu} = \frac{\hat{G}_N^{yx}}{\hat{G}_N^{ux}} = \frac{\hat{\Phi}_{yx}^N}{\hat{\Phi}_{ux}^N}. \tag{69}$$

For a discussion on "classical" joint input–output methods, see Gustavsson et al. (1977). Recently, several new prediction error methods that fall in this framework have been suggested, e.g., the two-stage method (Van den Hof and Schrama, 1993) and the related projection method (Forssell and Ljung, 1999, to appear), and the coprime factor identification scheme (Van den Hof et al., 1995). In the following, we will discuss how to implement the joint input–output approach utilizing the fact that the prediction error method allows arbitrary parameterizations of the models. This will allow us to do a unified analysis of all joint input–output methods using the general theory and to compare the results with the corresponding ones obtained for the direct and indirect approaches.

5.2. Joint input–output identification using the prediction error method

Suppose that the regulator is linear and of the form (25) and suppose e and d are independent noise sources (d does not necessarily have to be white). The output y and input u then obey

$$\begin{bmatrix} y(t) \\ u(t) \end{bmatrix} = \mathsf{G}_0(q)r(t) + \mathsf{H}_0(q)\begin{bmatrix} e(t) \\ d(t) \end{bmatrix}, \tag{70}$$

$$\mathsf{G}_0(q) = \begin{bmatrix} G_0^c(q) \\ S_0^i(q) \end{bmatrix},$$

$$\mathsf{H}_0(q) = \begin{bmatrix} S_0(q)H_0(q) & G_0(q)S_0^i(q) \\ -K(q)S_0(q)H_0(q) & S_0^i(q) \end{bmatrix}. \tag{71}$$

The idea is to identify the augmented system (70) using a model of the form

$$\begin{bmatrix} y(t) \\ u(t) \end{bmatrix} = \mathsf{G}(q,\theta)r(t) + \mathsf{H}(q,\theta)\begin{bmatrix} e(t) \\ d(t) \end{bmatrix}, \tag{72}$$

where the parameterizations of the transfer functions $\mathsf{G}(q,\theta)$ and $\mathsf{H}(q,\theta)$, for the time being, are not further specified, and to compute an open-loop estimate $\hat{G}_N$ using $\hat{\mathsf{G}}_N$ and $\hat{\mathsf{H}}_N$. Different parameterizations will lead to different methods, as we shall see. Here we have assumed a nonzero reference signal r. There are also methods that assume $r = 0$ and model y and u jointly as time series and retrieves the open-loop system from an estimate of H_0 (see, e.g., Gustavsson et al., 1977; Söderström and Stoica, 1989).

If we parameterize $G(q,\theta)$ in Eq. (72) as

$$\mathsf{G}(q,\theta) = \begin{bmatrix} G^{yr}(q,\theta) \\ G^{ur}(q,\theta) \end{bmatrix} \tag{73}$$

and estimate the parameters θ using, e.g., a fixed noise model $\mathsf{H}(q,\theta) = \mathsf{H}_*(q)$ we can use the straightforward approach of simply dividing the estimates $\hat{G}_N^{yr}$ and $\hat{G}_N^{ur}$ (cf. Eqs. (68) and (69)):

$$\hat{G}_N(q) = \hat{G}_N^{yr}(q)(\hat{G}_N^{ur}(q))^{-1}. \tag{74}$$

We can also include a prefilter for r and use the filtered signal $x = F_r r$ (again, cf. Eqs. (68) and (69)) in the model:

$$\begin{bmatrix} y(t) \\ u(t) \end{bmatrix} = \begin{bmatrix} G^{yx}(q,\theta) \\ G^{ux}(q,\theta) \end{bmatrix} x(t) + \mathsf{H}_*(q)\begin{bmatrix} e(t) \\ d(t) \end{bmatrix} \tag{75}$$

and use

$$\hat{G}_N(q) = \hat{G}_N^{yx}(q)(\hat{G}_N^{ux}(q))^{-1}. \tag{76}$$

It can be shown that the prefilter F_r can be used to shape the bias distribution of the resulting models (in case of undermodeling). See Corollary 7 below.

A drawback with proceeding bluntly as in Eq. (76) is that the estimate $\hat{G}_N$ inevitably will be of high order. This problem is similar to the one we are faced with in the indirect method, where we noted that solving for the open-loop estimate in Eq. (50) typically gives high-order estimates. A solution is to parameterize $G^{yx}(q,\theta)$ and $G^{ux}(q,\theta)$ as

$$G^{yx}(q,\theta) = G^{yu}(q,\rho)G^{ux}(q,\beta), \quad G^{ux}(q,\theta) = G^{ux}(q,\beta),$$

$$\theta = \begin{bmatrix} \rho \\ \beta \end{bmatrix}. \tag{77}$$

This would give us control over the order of $\hat{G}_N$ through the factor $G^{yu}(q, \rho)$, and circumvent the step (76). If we disregard the correlation between the noise sources affecting y and u, we may also first estimate the β-parameters separately using u and r and then estimate the ρ-parameters using y and r, keeping the β-parameters fixed to their estimated values. We will study such methods further in Section 5.3 below.

Another interesting idea is to use the prefilter F_r to tune the method to give models $\hat{G}_N^{yx}$ and $\hat{G}_N^{ux}$ that are normalized coprime ($\hat{G}_N^{yx}$, $\hat{G}_N^{yx}$ stable, $(\hat{G}_N^{yx})^* \hat{G}_N^{yx} + (\hat{G}_N^{ux})^* \hat{G}_N^{ux} = I$). If we succeed in estimating normalized coprime factors, the open-loop model $\hat{G}_N$ will be of minimal order (the McMillan degree of $\hat{G}_N$ will be equal to that of $\hat{G}_N^{yx}$ and $\hat{G}_N^{ux}$) and we will achieve a complete elimination of redundant dynamics. This is the idea behind the so-called coprime factor identification scheme (Van den Hof et al., 1995). (Note, though, that in practice $\hat{G}_N^{yx}$ and $\hat{G}_N^{ux}$ will never be exactly normalized coprime due to noise and the fact that the proper filter F_r to achieve normalized coprimeness cannot be calculated unless the system is perfectly known. In Van den Hof et al. (1995) it is therefore suggested to use an iterative procedure to achieve approximate normalized coprimeness.) In this method it is also suggested that the models $\hat{G}_N^{yx}$ and $\hat{G}_N^{ux}$ are parameterized on a common denominator form, similar to Eq. (77), to simplify the computation of $\hat{G}_N$.

5.3. The two-stage and projection methods

The two-stage method (Van den Hof and Schrama, 1993) and the projection method (Forssell and Ljung, 1999, to appear) can both be explained using the following two steps:

(1) Estimate the β-parameters in the model

$$u(t) = S(q, \beta)r(t) + H_1(q)e(t) \tag{78}$$

and construct the signal $\hat{u} = \hat{S}_N r$.

(2) Identify the open-loop system using the model

$$y(t) = G(q, \rho)\hat{u}(t) + H_2(q)e(t). \tag{79}$$

This is the same parameterization idea as in Eq. (77) and these methods can thus be seen as joint input–output methods where the correlation between the noise sources in Eqs. (78) and (79) is ignored. This classification will be further motivated in the next section.

The two-stage method was originally suggested as a method for consistently estimating the open-loop system, regardless of the noise model used, in case of linear feedback. Indeed, if the feedback is linear, given by Eq. (15), and the first step of the algorithm was successful so that $\hat{S}_N = S_0^i$, then the simulated signal $\hat{u}$ will be exactly equal to the noise-free part of the input and the open-loop system can be consistently estimated using Eq. (79). See Corollary 7 below. (Note that in the first step a high-order model of S_0 can be used, since we in the second step can control the open-loop model order independently. This implies that in most cases we can find a practically unbiased estimate of S_0^i in the first step.)

The two-stage method is quite robust and rather simple to use, therefore it is an attractive alternative to many of the existing closed-loop identification methods, including the direct method. However, the two-stage method will fail if the controller is nonlinear and/or contains some unknown disturbance signal d like in Eq. (25) that is not accounted for in the model. And, as we mentioned in the introduction, this will typically be the case in industrial applications even when the basic controller structure is linear. With the projection method these problems can be circumvented and, consequently, this method can be applied to systems with arbitrary feedback mechanisms, just as the direct method, and gives consistent estimates of G_0 regardless of the noise model used in the second step (79). The idea is to use a noncausal FIR model

$$u(t) = S(q, \beta)r(t) + e(t) = \sum_{k=-M_1}^{M_2} s_k r(t-k) + e(t) \tag{80}$$

with M_1 and M_2 chosen so large that any correlation between $u(t)$ and $r(s)$ for $t - s \notin [-M_1, \ldots, M_2]$ can be ignored, in the first step of the algorithm. This will imply that $\hat{u}$ and $\tilde{u} = u - \hat{u}$ (asymptotically) will be uncorrelated or, in a least-squares estimation language, *orthogonal* (hence the name *projection* method). As we shall see, this is the key to obtaining a consistent open-loop estimate.

It can thus be seen that the noncausal taps in the FIR model (80) is what distinguishes the projection method from the two-stage method, which considers only causal models. In Forssell and Ljung (1999, to appear) (see also below) it is shown that the noncausal taps cannot be excluded in general if the controller is nonlinear, even though the feedback system is causal (which we assume).

The model (80) can be seen as an approximation of the *Wiener smoother* (e.g., Kailath, 1981) solving the linear least mean square problem (assuming infinite data records):

$$\min_S \bar{E}(u(t) - S(q)r(t))^{\mathrm{T}}(u(t) - S(q)r(t)),$$
$$S(q) = \sum_{k=-\infty}^{\infty} s_k q^{-k}. \tag{81}$$

The Wiener smoother has the property that $\hat{u}\ (= \hat{S}r)$ and $\tilde{u}\ (= u - \hat{u})$ are uncorrelated. In the frequency domain the solution can be expressed as $\hat{S} = \Phi_{ur}(\Phi_r)^{-1}$ (cf. Eq. (68)). In the linear case this equals S_0^i which of course is causal operator. In the nonlinear case, however, the

asymptotically optimal Wiener smoother need not be causal even though the feedback system is causal. This is the reason why a noncausal model can be strictly better than a causal one (or, equivalently, why the projection method outperforms the two-stage method) asymptotically. In practice, with finite data records, the noncausal taps in Eq. (80) can be used to decrease the de facto correlation between $\hat{u}$ and $\tilde{u}$, which will have a positive effect on the quality of the final estimate. This is illustrated in the following example.

Example 4. Consider the closed-loop system in Fig. 2. The system G_0 is given by

$$G_0(q) = q^{-1} - 0.8q^{-2} \tag{82}$$

and the feedback controller is

$$\kappa(x) = \begin{cases} 0.25x + 4 & \text{if } x \geq 0, \\ 0.25x - 4 & \text{if } x < 0. \end{cases} \tag{83}$$

Thus, apart from just a proportional term, the controller consists of a columbic friction element.

In order to identify G_0 we performed a noise-free ($v = 0$) simulation of this system using a Gaussian white-noise reference signal: 5000 data samples were collected. Thus the only errors in the models should be bias errors.

To identify the system we used the two-stage and the projection methods. The following models were employed in the first step of the two-stage method:

(1) An FIR model:

$$u(t) = b_0 r(t) + b_1 r(t-1) + \cdots + b_{10} r(t-10) + e(t). \tag{84}$$

(2) An ARX model:

$$u(t) = -a_1 u(t-1) - \cdots - a_5 u(t-5) + b_0 r(t) + \cdots + b_5 r(t-5) + e(t). \tag{85}$$

In the first step of the projection method the model was

$$u(t) = s_{-5} r(t+5) + \cdots + s_0 r(t) + \cdots + s_5 r(t-5) + e(t). \tag{86}$$

That is, a noncausal FIR model. Notice that the same number of parameters are used in all three cases. In the second step of the methods the model was chosen as

$$y(t) = b_1 \hat{u}(t-1) + b_2 \hat{u}(t-2) + e(t), \tag{87}$$

where $\hat{u}$ was obtained using the first step models.

The identification results are summarized in Table 1. Here we can note that the two variants of the two-stage method give biased results while the projection method gives a very accurate estimate. Bode plots of the resulting model errors are shown in Fig. 3 together with a plot of G_0. The only practical difference between the two-stage method and the projection method is that the latter uses a noncausal model in the first step of the method. As we saw above, the noncausal taps in the FIR model used in the projection method are important for obtaining asymptotically unbiased results. This example shows that the noncausal taps can also make a difference in case of finite data records. Note especially that we in this example have used a Gaussian white noise reference signal and that the nonlinearity is static, so there should be no noncausal correlation between r and u (asymptotically). Despite this, the projection method still outperforms the two-stage method in this example. The

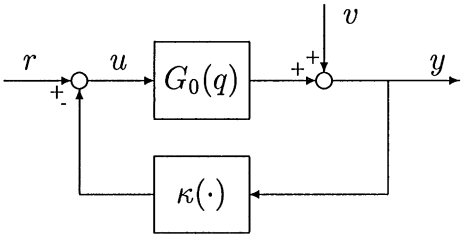


Fig. 2. Closed-loop system with nonlinear feedback.

Table 1
Summary of identification results.

	b_1	b_2
True system	1.0000	−0.8000
Two-stage (FIR)	0.9595	−0.8258
Two-stage (ARX)	0.8857	−0.6906
Projection	0.9985	−0.7965

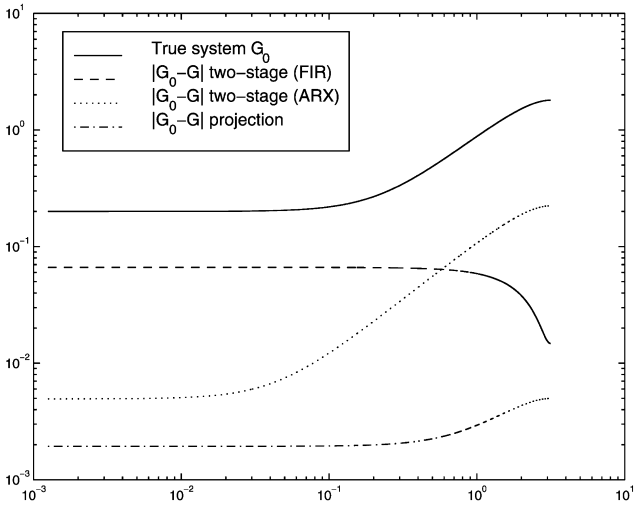


Fig. 3. Bode plot of the true system and the errors in the estimated models.

reason is that there is in fact a significant noncausal correlation between r and u, in this finite sample case, that is picked up by the noncausal FIR model.

Finally, it can be noted that the direct method, applied with a correct/parameterized noise model, will retrieve the true system perfectly regardless of the controller, since the simulation is noise-free. The indirect method is not easy to apply in this kind of situation, even if the controller is perfectly known. The reason is that the nonlinearity will make the estimation problem non-trivial. If we ignore the nonlinearity when applying the indirect method the resulting estimate will (quite naturally) be very poor.

5.4. Unifying framework for all joint input–output methods

It will now be shown that the joint input–output approach can be seen as a "direct" approach where both the system and the controller are identified and that the two-stage and projection methods are indeed joint input–output methods with specially chosen noise models. To achieve this we will exploit the links between the prediction error method (30)–(32) and maximum likelihood estimation.

Consider the joint system (70) and assume that d is white-noise with covariance matrix Λ_d, independent of e. The maximum likelihood estimates of G_0 and H_0 are computed as

$$\min_{\theta\in D_{\mathscr{M}}} \frac{1}{N}\sum_{t=1}^{N}\begin{bmatrix} y(t)-\hat{y}(t|\theta)\\ u(t)-\hat{u}(t|\theta)\end{bmatrix}^{\mathrm{T}}\begin{bmatrix}\Lambda_0 & 0\\ 0 & \Lambda_d\end{bmatrix}^{-1}\begin{bmatrix} y(t)-\hat{y}(t|\theta)\\ u(t)-\hat{u}(t|\theta)\end{bmatrix}, \tag{88}$$

where

$$\begin{bmatrix}\hat{y}(t|\theta)\\ \hat{u}(t|\theta)\end{bmatrix} = \mathsf{H}^{-1}(q,\theta)\mathsf{G}(q,\theta)r(t) + (I-\mathsf{H}^{-1}(q,\theta))\begin{bmatrix} y(t)\\ u(t)\end{bmatrix}. \tag{89}$$

The parameterizations of G and H can be arbitrary. The relation (70) was obtained using the assumption that the noise e affects the open-loop system only and the disturbance d affects the regulator only. The natural way to parameterize H in order to reflect these assumptions in the model is

$$\mathsf{H}(q,\theta)=\begin{bmatrix} (I+\mathsf{G}(q,\theta)K(q,\theta))^{-1}H(q,\theta) & G(q,\theta)(I+K(q,\theta)G(q,\theta))^{-1}\\ -K(q,\theta)(I+G(q,\theta)K(q,\theta))^{-1}H(q,\theta) & (I+K(q,\theta)G(q,\theta))^{-1}\end{bmatrix}. \tag{90}$$

With G parameterized as

$$\mathsf{G}(q,\theta)=\begin{bmatrix} G(q,\theta)(I+K(q,\theta)G(q,\theta))^{-1}\\ (I+K(q,\theta)G(q,\theta))^{-1}\end{bmatrix}, \tag{91}$$

we get

$$\begin{bmatrix}\hat{y}(t|\theta)\\ \hat{u}(t|\theta)\end{bmatrix} = \begin{bmatrix}0\\ I\end{bmatrix}r(t) + \begin{bmatrix} I-H^{-1}(q,\theta) & H^{-1}(q,\theta)G(q,\theta)\\ -K(q,\theta) & 0\end{bmatrix}\begin{bmatrix} y(t)\\ u(t)\end{bmatrix} \tag{92}$$

or

$$\hat{y}(t|\theta) = (I-H^{-1}(q,\theta))y(t) + H^{-1}(q,\theta)G(q,\theta)u(t), \tag{93}$$

$$\hat{u}(t|\theta) = r(t) - K(q,\theta)y(t). \tag{94}$$

The predictor (93) is the same as for the direct method (cf. Eq. (28)), while Eq. (94) is the natural predictor for estimating the regulator K. The maximum likelihood estimate becomes

$$\min_{\theta\in D_{\mathscr{M}}}\frac{1}{N}\left\{\sum_{t=1}^{N}(y(t)-\hat{y}(t|\theta))^{\mathrm{T}}\Lambda_0^{-1}(y(t)-\hat{y}(t|\theta)) + \sum_{t=1}^{N}(u(t)-\hat{u}(t|\theta))^{\mathrm{T}}\Lambda_d^{-1}(u(t)-\hat{u}(t|\theta))\right\}. \tag{95}$$

This shows that we may view the joint input–output method as a combination of direct identification of the open-loop system and a direct identification of the regulator. Note that this holds even in the case where $r(t)=0$.

If the parameterization of the regulator K is independent of the one of the system the two terms in Eq. (95) can be minimized separately which decouples the two identification problems. This idea is used in the two-stage and projection methods to be studied next.

Let us return to Eq. (70). Suppose that the model (cf. Eq. (77) and Eqs. (78)–(79))

$$\begin{bmatrix}\hat{y}(t|\theta)\\ \hat{u}(t|\theta)\end{bmatrix} = \begin{bmatrix} G(q,\rho)\\ I\end{bmatrix}S^{\mathrm{i}}(q,\beta)r(t), \quad \theta=\begin{bmatrix}\rho\\ \beta\end{bmatrix} \tag{96}$$

is used. The maximum likelihood estimate becomes

$$\min_{\theta\in D_{\mathscr{M}}}\frac{1}{N}\left\{\sum_{t=1}^{N}(y(t)-G(q,\rho)S^{\mathrm{i}}(q,\beta)r(t))^{\mathrm{T}}\Lambda_0^{-1}\times(y(t)-G(q,\rho)S^{\mathrm{i}}(q,\beta)r(t)) + \sum_{t=1}^{N}(u(t)-S^{\mathrm{i}}(q,\beta)r(t))^{\mathrm{T}}\Lambda_d^{-1}\times(u(t)-S^{\mathrm{i}}(q,\beta)r(t))\right\}. \tag{97}$$

It will be assumed that $\Lambda_0=\lambda_0\cdot I$ and $\Lambda_d=\lambda_d\cdot I$. (If Λ_0 and Λ_d are known this can always be achieved by suitable prefiltering or, equivalently, by changing the noise models.) As λ_d tends to zero the maximum likelihood estimate (97) will approach the one obtained with

the two-stage/projection methods. This is true because for small λ_d the β-parameters will minimize

$$\frac{1}{N}\sum_{t=1}^{N}(u(t)-S^{\mathrm{i}}(q,\beta)r(t))^{\mathrm{T}}(u(t)-S^{\mathrm{i}}(q,\beta)r(t)) \tag{98}$$

regardless of the ρ-parameters, which then will minimize

$$\frac{1}{N}\sum_{t=1}^{N}(y(t)-G(q,\rho)S^{\mathrm{i}}(q,\beta)r(t))^{\mathrm{T}}(y(t) - G(q,\rho)S^{\mathrm{i}}(q,\beta)r(t)). \tag{99}$$

Thus the two-stage and projection methods may be viewed as special cases of the general joint input–output approach corresponding to special choices of the noise models. This means especially that any results that hold for the joint input–output approach without constraints on the noise models, hold for the two-stage and projection methods as well. We will use this fact in Corollary 9 below.

6. Convergence results for the closed-loop identification methods

Let us now apply the result of Theorem A.1 to the special case of closed-loop identification. We will first study the direct approach. (In the following we will write G_0 as short for $G_0(\mathrm{e}^{\mathrm{i}\omega})$ and G_θ as short for $G(\mathrm{e}^{\mathrm{i}\omega},\theta)$, etc. The subscript θ is included to emphasize the parameter dependence.)

6.1. The direct approach

Corollary 5. *Consider the situation in Theorem A.1 and suppose that u is persistently exciting. With a model structure*

$$y(t)=G(q,\theta)u(t)+H(q,\theta)e(t), \tag{100}$$

we then have that, w. p. 1 as $N\to\infty$,

$$\hat{\theta}_N\to D_{\mathrm{c}}=\arg\min_{\theta\in D_{\mathscr{M}}}\int_{-\pi}^{\pi}\mathrm{tr}[\{(G_0+B_\theta-G_\theta)\Phi_u \times(G_0+B_\theta-G_\theta)^* + (H_0-H_\theta)\Phi_e^r(H_0-H_\theta)^*\}(H_\theta\Lambda H_\theta^*)^{-1}]\,\mathrm{d}\omega, \tag{101}$$

where

$$B_\theta=(H_0-H_\theta)\Phi_{eu}\Phi_u^{-1}. \tag{102}$$

The proof is given in Appendix B.1.

Remark. The "bias-pull" term B_θ defined in Eq. (102) can be noncausal depending, e.g., on the nature of the feedback. However, since both G_0 and G_θ are causal we can replace B_θ by a causal counterpart in Eq. (101). The exact statement is as follows. If we let $[A]_+$ denote the causal part and $[A]_-$ the anti-causal part of A we can reformulate Eq. (101) as

$$\hat{\theta}_N\to D_{\mathrm{c}}=\arg\min_{\theta\in D_{\mathscr{M}}}\int_{-\pi}^{\pi}\mathrm{tr}[\{(G_0+B_\theta^+-G_\theta)\Phi_u \times(G_0+B_\theta^+-G_\theta)^* + (H_0-H_\theta)\Phi_e^r \times(H_0-H_\theta)^* + B_\theta^-\Phi_u(B_\theta^-)^*\} \times(H_\theta\Lambda H_\theta^*)^{-1}]\,\mathrm{d}\omega, \tag{103}$$

where

$$B_\theta^+=H_\theta[H_\theta^{-1}(H_0-H_\theta)\Phi_{eu}M_u^{-*}]_+M_u^{-1}, \tag{104}$$

$$B_\theta^-=H_\theta[H_\theta^{-1}(H_0-H_\theta)\Phi_{eu}M_u^{-*}]_-M_u^{-1}. \tag{105}$$

A proof is included in Appendix B.1.

If the parameterization of the model G and the noise model H is flexible enough so that $\mathscr{S}\in\mathscr{M}$ then $\bar{V}(\theta_0)=\Lambda_0$, and provided this is a unique minimum we thus have $D_{\mathrm{c}}=\{\theta_0\}$. See, e.g., the discussion following Eq. (39) and Theorem 8.3 in Ljung (1987).

We can also note that if the system operates in open loop, so that $\Phi_{ue}=0$, then the bias term $B_\theta=0$ regardless of the noise model H_θ and the limit model will be a compromise between making G_θ as an good approximation of G_0 as possible and making H_θ as an good approximation of H_0 as possible (with the indicated frequency weightings). A consequence of this is that in open loop we can use fixed noise models and still get consistent estimates of G_0 provided that $G_0\in\mathscr{G}$ and certain identifiability conditions hold. See, e.g., Theorem 8.4 in Ljung (1987).

In case of undermodeling we get a characterization of the bias error through the term B_θ in Eq. (102). A straightforward upper bound on $\bar{\sigma}(B_\theta)$ (the maximal singular value of B_θ) is

$$\bar{\sigma}(B_\theta)\le\bar{\sigma}(H_0-H_\theta)\bar{\sigma}(\Phi_{eu})\frac{1}{\underline{\sigma}(\Phi_u)}. \tag{106}$$

($\underline{\sigma}(\Phi_u)$ denotes the minimal singular values of Φ_u.) If the feedback is linear and given by Eq. (15) we get the following alternative upper bound:

$$\bar{\sigma}(B_\theta)\le\bar{\sigma}(H_0-H_\theta)\sqrt{\frac{\bar{\sigma}(\Lambda_0)}{\underline{\sigma}(\Phi_u)}}\sqrt{\frac{\bar{\sigma}(\Phi_u^e)}{\underline{\sigma}(\Phi_u)}}. \tag{107}$$

It follows that the bias inclination in the G-estimate will be small in frequency ranges where either (or all) of the following holds:

- The noise model is good ($\bar{\sigma}(H_0-H_\theta)$ is small).
- The feedback noise contribution to the input spectrum ($\bar{\sigma}(\Phi_u^e)/\underline{\sigma}(\Phi_u)$) is small.
- The signal-to-noise ratio is high ($\bar{\sigma}(\Lambda_0)/\underline{\sigma}(\Phi_u)$ is small).

In particular, if a reasonable flexible, independently parameterized noise model is used the bias inclination of the G-estimate can be small.

6.2. The indirect approach

We will now give the corresponding convergence result for the indirect method. Here we assume that this method is implemented using a model structure of the form (49) where G^c is parameterized as in Eq. (52).

Corollary 6. *Consider the situation in Theorem A.1 and suppose that r is persistently exciting and that the input u is given by Eq. (25). Then, if the model structure is*

$$y(t) = G^c(q,\theta)r(t) + H_*(q)e(t), \tag{108}$$

we have that, w. p. 1 as $N \to \infty$,

$$\hat{\theta}_N \to D_c = \arg\min_{\theta\in D_{\mathcal{M}}} \int_{-\pi}^{\pi} \mathrm{tr}\big[(G_0^c + \bar{B} - G_\theta^c)\Phi_r \times (G_0^c + \bar{B} - G_\theta^c)^*(H_*\Lambda H_*^*)^{-1}\big]\,\mathrm{d}\omega, \tag{109}$$

where

$$\bar{B} = G_0^c\Phi_{dr}\Phi_r^{-1}. \tag{110}$$

The proof is given in Appendix B.2.

Remark. Since we consider causal systems and models we can alternatively write Eq. (109) as

$$\hat{\theta}_N \to D_c = \arg\min_{\theta\in D_{\mathcal{M}}} \int_{-\pi}^{\pi} \mathrm{tr}\big[(G_0^c + \bar{B}^+ - G_\theta^c)\Phi_r \times (G_0^c + \bar{B}^+ - G_\theta^c)^*(H_*\Lambda H_*^*)^{-1}\big]\,\mathrm{d}\omega, \tag{111}$$

where

$$\bar{B}^+ = H_*\big[H_*^{-1}G_0^c\Phi_{dr}M_r^{-*}\big]_+ M_r^{-1}. \tag{112}$$

This is shown in Appendix B.2.

From Eq. (109) we can draw a number of conclusions regarding the limit estimates for the indirect method. We remember that the closed-loop model G^c is parameterized as $G^c(q,\theta) = (I + G(q,\theta)K(q))^{-1}G(q,\theta)$.

If d is uncorrelated with r then $\Phi_{dr} = 0$ and $\bar{B} = 0$ (cf. Eq. (110)), and the G-estimate will try to minimize the norm of

$$G_0^c - G_\theta^c = G_0^c - S_\theta G_\theta = S_0(G_0 - G_\theta)S_\theta^i \tag{113}$$

with the indicated frequency weighting. This shows that the indirect method can give consistent estimates of G_0 if the parameterization of G is flexible enough, that is, if $G_0 \in \mathcal{G}$, even with a fixed noise model. This is the main advantage of the indirect method. In case of undermodelling the resulting G-estimate will try to minimize the mismatch between G_0 and G and at the same time try to minimize the model sensitivity function S^i. There will thus be a "bias-pull" towards transfer functions that give a small sensitivity for the given regulator, which can be advantageous, e.g., if the model is to be used for control (Gevers, 1993; Van den Hof and Schrama, 1995). Unlike Eq. (101) it is not easy to quantify this bias error in the open-loop model, though.

If d and r are correlated then $\bar{B} \neq 0$ and the G-estimate will try to minimize the (frequency weighted) norm of

$$G_0^c + \bar{B} - (I + G_\theta K)^{-1}G_\theta \tag{114}$$

which can lead to an arbitrarily bad open-loop estimate depending on $\bar{B}$. This situation will, e.g., occur if the assumed controller parameters deviate from the true ones or if the true controller is nonlinear. See the discussion in Section 2. (In case we know u in addition to y and r there is a simple remedy for this problem, since then we can construct $r' = u + Ky\,(= r + d)$ and use this signal instead of r which would eliminate the bias.)

6.3. The joint input–output approach

Finally, we will treat the joint input–output approach. Here we will study two cases, first a "classical" joint input–output method, where the final G estimate is found as in Eqs. (72)–(74), and, second, the two-stage/projection method.

Corollary 7. *Consider the situation in Theorem A.1 and suppose that r is persistently exciting and that the input u* is given by *Eq. (25). Then*

(1) *if the model structure is*

$$\begin{bmatrix} y(t) \\ u(t) \end{bmatrix} = \begin{bmatrix} G^c(q,\theta) \\ S^i(q,\theta) \end{bmatrix} r(t) + \mathsf{H}_*(q)\begin{bmatrix} e(t) \\ d(t) \end{bmatrix} \tag{115}$$

we have that, w. p. 1 as $N \to \infty$,

$$\hat{\theta}_N \to D_c = \arg\min_{\theta\in D_{\mathcal{M}}} \int_{-\pi}^{\pi} \mathrm{tr}\left[\begin{bmatrix} G_0^c\cdot D - G^c(q,\theta) \\ S_0^i\cdot D - S^i(q,\theta) \end{bmatrix} \times \Phi_r \begin{bmatrix} G_0^c\cdot D - G^c(q,\theta) \\ S_0^i\cdot D - S^i(q,\theta) \end{bmatrix}^* \times (\mathsf{H}_*\Lambda\mathsf{H}_*^*)^{-1}\right]\mathrm{d}\omega, \tag{116}$$

where

$$D = I + \Phi_{dr}\Phi_r^{-1}, \tag{117}$$

(2) *if the model structure is*

$$y(t) = G(q,\theta)\hat{u}(t) + H_*(q)e(t), \quad \hat{u}(t) = \hat{S}_N(q)r(t), \quad \tilde{u}(t) = u(t) - \hat{u}(t) \tag{118}$$

we have that, w. p. 1 *as* $N \to \infty$,

$$\hat{\theta}_N \to D_c = \arg\min_{\theta \in D_{\mathcal{M}}} \int_{-\pi}^{\pi} \mathrm{tr}[(G_0 + \tilde{B} - G_\theta)\Phi_{\hat{u}} \times (G_0 + \tilde{B} - G_\theta)^*(H_*\Lambda H_*^*)^{-1}]\,\mathrm{d}\omega, \tag{119}$$

where

$$\tilde{B} = G_0\Phi_{\tilde{u}\hat{u}}\Phi_{\hat{u}}^{-1}. \tag{120}$$

The proof is given in Appendix B.3.

Remark. We could also rewrite Eqs. (116) and (118) in similar forms as Eq. (111) but for brevity we do not give the details here.

Let us first comment on the result (116). We assume that the open-loop estimate is computed as $\hat{G}_N = \hat{G}_N^c(\hat{S}_N^i)^{-1}$. If $D = I$ and the parameterizations of G^c and S^i are flexible enough, then this method will give consistent estimates of G_0, since $\hat{G}_N = \hat{G}_N^c(\hat{S}_N^i)^{-1} \to G_0^c(S_0^i)^{-1} = G_0$ as N tends to infinity, regardless of the noise models used, just as for the indirect method. An advantage compared to the indirect method is that the controller need not be known. We can also note that if the parameterizations of G^c and S^i are such that $\hat{G}_N^c \to G_0^c D$ and $\hat{S}_N^i \to S_0^i D$ as N tends to infinity, then this joint input–output method gives consistent G-estimates regardless of the disturbance d. This follows since $G_0^c D(S_0^i D)^{-1} = G_0$ regardless of D.

The result (119) holds for both the two-stage and the projection method. It is clear that any correlation between $\hat{u}$ and $\tilde{u}$ will deteriorate the G-estimate in these methods. The error is quantified by the term $\tilde{B}$ defined in Eq. (120). With the projection method $\Phi_{\tilde{u}\hat{u}} = 0$ is (asymptotically) achieved, by construction, which implies that $\tilde{B} = 0$. Hence, with this method it is possible to obtain consistent estimates of G_0 with fixed noise models (prefilters), regardless of the feedback just as with the direct method. An important advantage compared to the direct method is that with the projection method we can guarantee consistency even if a fixed noise model is used. This also means that in case of undermodeling the model can be fit to the data with arbitrary frequency weighting with this method.

7. Asymptotic variance of black-box transfer function estimate

Consider first the direct approach. The following result follows directly from Theorem A.2 after noting that

$$\Phi_{\chi_0}^{-1} = \begin{bmatrix} (\Phi_u^r)^{-1} & -(\Phi_u^r)^{-1}\Phi_{ue}\Lambda_0^{-1} \\ -\Lambda_0^{-1}\Phi_{eu}(\Phi_u^r)^{-1} & (\Phi_e^r)^{-1} \end{bmatrix}. \tag{121}$$

Corollary 8. *Suppose that the conditions in Theorem A.2 hold. Then*

$$\sqrt{N}\,\mathrm{vec}[\hat{G}_N(e^{i\omega}, n, \delta) - G_n^*(e^{i\omega})] \in AsN(0, P(\omega, n, \delta)) \quad \text{as } N \to \infty \text{ for fixed } n,\ \delta, \tag{122}$$

where

$$\lim_{\delta \to 0}\lim_{n \to \infty} \frac{1}{n} P(\omega, n, \delta) = (\Phi_u^r(\omega))^{-T} \otimes \Phi_v(\omega) \tag{123}$$

(The notation is explained in Appendix A.2.)

An intuitive, but not formally correct, interpretation of Eqs. (122) and (123) is

$$\mathrm{Cov}\,\mathrm{vec}[\hat{G}_N(e^{i\omega})] \sim \frac{n}{N}(\Phi_u^r(\omega))^{-T} \otimes \Phi_v(\omega) \tag{124}$$

as the model order n as well as the number of data N tends to infinity.

The difference between Eq. (124) (in the sequel we will use Eq. (124) when referring to the result in Corollary 8) and the corresponding open-loop expression (44) is thus that Φ_u^r replaces Φ_u, or, in other words, that the part of the input spectrum that originates from the reference signal r replaces the total input spectrum.

The expression (124) — which also is the asymptotic Cramèr–Rao lower limit — tells us precisely "the value of information" of closed-loop experiments. It is the noise-to-signal ratio (where "signal" is what derives from the injected reference) that determines how well the open-loop transfer function can be estimated. From this perspective, that part of the input that originates from the feedback noise has no information value when estimating G.

If we in the direct method use a fixed noise model equal to the true one, $H(q, \theta) = H_*(q) = H_0(q)$, we have that

$$\mathrm{Cov}\,\mathrm{vec}[\hat{G}_N(e^{i\omega}, n)] \sim \frac{n}{N}(\Phi_u(\omega))^{-T} \otimes \Phi_v(\omega) \tag{125}$$

as the model order tends to infinity, just as in the open-loop case. This is true since the covariance matrix is determined entirely by the second-order properties (the spectrum) of the input, and it is immaterial whether this spectrum is a result of open-loop or closed-loop operation. We emphasize that this alternative result holds only if we use a *fixed* and correct noise model. The importance of this for the improved accuracy will be further discussed below.

It should be noted that Eq. (124) holds for the direct method if $\mathcal{S} \in \mathcal{M}$. If we restrict to linear feedback as in Eq. (15) we have $\Phi_u^r = S_0^i\Phi_r(S_0^i)^*$. In this case Eq. (124) also holds for the indirect and joint input–output approaches if $G_0 \in \mathcal{G}$, regardless of the

parameterizations and the noise models used. This follows from

Corollary 9. *Suppose that the input is generated as in Eq.* (15). *If the estimate* $\hat{G}_N$ *is found using either the indirect or the joint input–output approach and the conditions in Corollary A.*3 *hold. Then Eqs.* (122) *and* (123) *hold also for these methods.*

The proof is given in Appendix B.4.

We stress that in Corollary 9 we restricted to linear feedback. Hence, Φ_u^r in Eq. (123) should really be replaced by $\Phi_u^r = S_0^{\mathrm{i}}\Phi_r(S_0^{\mathrm{i}})^*$ in this case.

It can also be noted that the results in, for example, Eqs. (43), (44), and (124) are independent of the parameterization and an important assumption is that the model order tends to infinity. It is then quite natural that the differences between the closed-loop identification methods disappear, since, as we have seen, they may be viewed as different parameterizations of the same general prediction error method.

8. Asymptotic distribution of parameter vector estimates in the case of closed-loop data

8.1. The direct and indirect approaches

We will first consider the case when the dynamics and noise models are independently parameterized. The following two corollaries to Theorem A.4 in Appendix A.3 states what the asymptotic distribution of the parameter estimates will be for the direct and indirect approaches, respectively, in their standard formulations. The results will be given for the SISO case only and in the following G'_ρ will be used as short for $(\mathrm{d}/\mathrm{d}\rho)G(\mathrm{e}^{\mathrm{i}\omega}, \rho)|_{\rho=\rho_0}$, etc. λ_0 denotes the variance of e in the SISO case.

Corollary 10. *Assume that the conditions in Theorem A.*4 *hold and that the model structure* $\mathscr{M}$ *is*

$$y(t) = G(q, \rho)u(t) + H(q, \beta)e(t), \qquad \theta = \begin{bmatrix} \rho \\ \beta \end{bmatrix}. \tag{126}$$

Suppose that $\mathscr{S} \in \mathscr{M}$ *and that* $\mathscr{M}$ *is globally identifiable at* θ_0. *Suppose also that the regulator is given by Eq.* (15) *and that the data set* Z^∞ *is informative enough with respect to* $\mathscr{M}$. *Then*

$$\sqrt{N}(\hat{\rho}_N - \rho_0) \in AsN(0, P_\rho^{\mathrm{D}}), \tag{127a}$$

where

$$P_\rho^{\mathrm{D}} = \lambda_0(R_\rho^r + \Delta)^{-1}, \tag{127b}$$

$$R_\rho^r = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{\Phi_u^r}{|H_0|^2}\, G'_\rho G'^*_\rho \,\mathrm{d}\omega, \tag{127c}$$

$$\Delta = R_\rho^e - R_{\rho\beta}^e (R_\beta^e)^{-1} R_{\beta\rho}^e, \tag{127d}$$

$$R_\rho^e = \frac{1}{2\pi}\int_{-\pi}^{\pi} \lambda_0 |K|^2 |S_0|^2 G'_\rho G'^*_\rho \,\mathrm{d}\omega, \tag{127e}$$

$$R_{\rho\beta}^e = -\frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{\lambda_0}{|H_0|^2}\, KS_0 H_0 G'_\rho H'^*_\beta \,\mathrm{d}\omega, \tag{127f}$$

$$R_\beta^e = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{\lambda_0}{|H_0|^2}\, H'_\beta H'^*_\beta \,\mathrm{d}\omega. \tag{127g}$$

Furthermore,

$$0 \le \Delta \le R_\rho^e \tag{128}$$

and Δ *approaches zero as the order of the noise model tends to infinity, while* $\Delta = R_\rho^e$ *if the noise model is fixed and equal to the true one, that is, if* $H(q, \beta) = H_*(q) = H_0(q)$.

The proof is given in Appendix B.5.

Corollary 11. *Consider the same situation as in Corollary* 10 *but assume that the model structure is*

$$y(t) = \frac{G(q, \rho)}{1 + G(q, \rho)K(q)}\, r(t) + H^{\mathrm{c}}(q, \beta)e(t), \qquad \theta = \begin{bmatrix} \rho \\ \beta \end{bmatrix}. \tag{129}$$

Then

$$\sqrt{N}(\hat{\rho}_N - \rho_0) \in AsN(0, P_\rho^{\mathrm{I}}), \tag{130a}$$

where

$$P_\rho^{\mathrm{I}} = \lambda_0(R_\rho^r)^{-1} \tag{130b}$$

with R_ρ^r *given by Eq.* (127c).

If the noise model in Eq. (129) *is fixed* $H^{\mathrm{c}}(q, \beta) = H_*(q)$ *and the model structure now satisfies* $G_0 \in \mathscr{G}$ *and is globally identifiable at* ρ_0, *we have that*

$$\sqrt{N}(\hat{\rho}_N - \rho_0) \in AsN(0, \bar{P}_\rho^{\mathrm{I}}), \tag{131a}$$

where

$$\bar{P}_\rho^{\mathrm{I}} = R^{-1} Q R^{-1}, \tag{131b}$$

$$R = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{|S_0|^2 \Phi_u^r}{|H_*|^2}\, G'_\rho G'^*_\rho \,\mathrm{d}\omega, \tag{131c}$$

$$Q = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{|S_0|^4 \Phi_v \Phi_u^r}{|H_*|^4}\, G'_\rho G'^*_\rho \,\mathrm{d}\omega. \tag{131d}$$

Proof. The main result is a consequence of Theorem A.4. The expressions for the covariance matrices follow from the corresponding open-loop expressions (see, e.g., Eq. (9.55) in Ljung, 1987) after noting that $(G_\rho^{\mathrm{c}})' = S_0^2 G'_\rho$ and $H_0^{\mathrm{c}} = S_0 H_0$. □

An important observation regarding the result (127) is that the term Δ is entirely due to the noise part of the

input spectrum and since $\Delta \geq 0$ this contribution has a positive effect on the accuracy, contrarily to what one might have guessed. We conclude that in the direct method the noise in the loop is utilized in reducing the variance. For the indirect methods this contribution is zero as can be seen from Eq. (130). The reason for this is that in the indirect method only that part of the input that originates from the reference signal is used in the identification.

From Eq. (127) it is also clear that — from the accuracy point of view — the worst-case experimental conditions with respect to the reference signal is when there is *no* external reference signal present, that is, when $\Phi_r = 0$. In this case

$$\operatorname{Cov} \hat{\rho}_N \sim \frac{\lambda_0}{N} \Delta^{-1}. \tag{132}$$

Thus Δ characterizes the lower limit of achievable accuracy for the direct method. Now, if Δ is nonsingular, Eq. (127) says that we can consistently estimate the system parameters using the direct method even though no reference signal is present. The exact conditions for this to happen are given in Söderström et al. (1975) for some common special cases. However, even if Δ is singular it will have a beneficial effect on the variance of the estimates, according to Eq. (127). Only when $\Delta = 0$ there is no positive effect from the noise source on the accuracy of the estimates. According to Corollary 10 this would be the case when the order of the noise model tends to infinity. At the other extreme we have that a fixed (and correct) noise model will make Δ as "large" as it can be.

This puts the finger on the value of information in the noise source e for estimating the dynamics:

It is the knowledge/assumption of a constrained noise model that improves the estimate of G_0.

This also explains the difference between Eq. (124) (which assumes the noise model order to tend to infinity) and Eq. (125) (which assumes a fixed and correct noise model).

For the indirect approach, we can further note that for all fixed H_*,

$$P_\rho^{\mathrm{I}} \leq \bar{P}_\rho^{\mathrm{I}} \tag{133}$$

with equality for $H_* = H_0^{\mathrm{o}} = S_0 H_0$. By comparing Eq. (130) with Eq. (127) we also see that the covariance matrix for the indirect method will always be larger than the one for the direct. The difference is the term Δ that is missing in Eq. (130). We thus have the following ranking:

$$P_\rho^{\mathrm{D}} \leq P_\rho^{\mathrm{I}} \leq \bar{P}_\rho^{\mathrm{I}}. \tag{134}$$

8.2. Further results for the indirect approach

In Corollary 11 we considered two particular variants of the indirect approach. It is important to realize that the conclusions drawn in the previous section do not hold for all possible variants and parameterizations of the indirect approach. In this section we will study two other variants where first the closed-loop system is identified and then the open-loop system is estimated by solving an over-determined system of equations, relating the open-loop parameters to the closed-loop parameters, in a least squares sense using the knowledge of the controller (cf. the discussion in Section 4.2). We will then see that it is possible to get the same level of accuracy with the indirect method as with the direct, even with finite model orders. This was first shown in Söderström et al. (1974) and in this section we will review and extend the results there. See also Gustavsson et al. (1977) and Söderström et al. (1991).

8.2.1. Indirect identification using ARMAX models

Let us now outline the idea. Suppose that the regulator is linear and of the form Eq. (15), that is

$$u(t) = r(t) - K(q)y(t). \tag{135}$$

Factorize K as

$$K(q) = \frac{X(q)}{Y(q)}, \tag{136}$$

where the polynomials X and Y are coprime. Let the closed- and open-loop model structures be

$$A^{\mathrm{c}}(q)y(t) = B^{\mathrm{c}}(q)r(t) + C^{\mathrm{c}}(q)e(t) \tag{137}$$

and

$$A(q)y(t) = B(q)u(t) + C(q)e(t), \tag{138}$$

respectively. From

$$G^{\mathrm{c}}(q) = \frac{G(q)}{1 + G(q)K(q)} \quad \text{and} \quad H^{\mathrm{c}}(q) = \frac{H(q)}{1 + G(q)K(q)} \tag{139}$$

(cf. Eq. (21)) it follows that we may solve for A, B and C in

$$\begin{aligned} A^{\mathrm{c}}(q) &= A(q)Y(q) + B(q)X(q), \\ B^{\mathrm{c}}(q) &= B(q)Y(q), \\ C^{\mathrm{c}}(q) &= C(q)Y(q). \end{aligned} \tag{140}$$

Let η and θ denote the closed- and open-loop parameter vectors, respectively. Then, for given η, Eq. (140) is a system of linear equations in θ from which θ can be estimated. If the Markov estimate is used, that is, if Eq. (140) is solved in a least-squares sense, using the estimated

covariances of $\hat{\eta}_N$ as weights, Corollary 12 below states that this indirect method is consistent under reasonable assumptions and gives the same accuracy as the direct method with an ARMAX model structure.

Corollary 12. *Consider the situation in Theorem A.4 and let the model structure $\mathcal{M}$ be given by Eq. (137). Assume that $\mathcal{M}$ is identifiable at η_0 and that the data set $Z^\infty = \{y(1), r(1), \dots, y(\infty), r(\infty)\}$ is informative enough with respect to $\mathcal{M}$. Assume also that $\mathcal{S} \in \mathcal{M}$ holds for the model structure (138). Then, if the open-loop parameters θ are estimated from Eq. (140) using the Markov estimate, we have that*

$$\sqrt{N}(\hat{\theta}_N - \theta_0) \in AsN(0, P_\theta^{\mathrm{AX}}), \tag{141a}$$

where

$$P_\theta^{\mathrm{AX}} = \lambda_0 [\bar{E}\psi(t, \theta_0)\psi^{\mathrm{T}}(t, \theta_0)]^{-1} \tag{141b}$$

$$\psi(t, \theta) = -\frac{\mathrm{d}}{\mathrm{d}\theta}\frac{A(q)}{C(q)}\left(y(t) - \frac{B(q)}{A(q)}u(t)\right). \tag{141c}$$

The proof is given in Appendix B.6.

We emphasize that Eq. (141) is exactly what we would get with the direct method applied using an ARMAX model. Through this corollary we have therefore established yet another link between the direct and indirect approaches. As will become more apparent below, the fact that the dynamics *model* and the noise *model* share the same poles is crucial for obtaining the same accuracy as with the direct approach.

8.2.2. Indirect identification using Box–Jenkins models

Suppose now that we in the first step in the indirect method use the following model structure with an independently parameterized noise model:

$$\begin{aligned} y(t) &= G^{\mathrm{c}}(q, \rho_\eta)r(t) + H^{\mathrm{c}}(q, \beta_\eta)e(t) \\ &= \frac{B^{\mathrm{c}}(q)}{F^{\mathrm{c}}(q)}r(t) + \frac{C^{\mathrm{c}}(q)}{D^{\mathrm{c}}(q)}e(t). \end{aligned} \tag{142}$$

This is also known as a Box–Jenkins model structure. Let the open-loop model be

$$\begin{aligned} y(t) &= G(q, \rho)u(t) + H(q, \beta)e(t) \\ &= \frac{B(q)}{F(q)}r(t) + \frac{C(q)}{D(q)}e(t). \end{aligned} \tag{143}$$

To find the indirect estimate of G_0 we may solve for B and F in (cf. Eq. (140))

$$\begin{aligned} B^{\mathrm{c}}(q) &= B(q)Y(q), \\ F^{\mathrm{c}}(q) &= F(q)Y(q) + B(q)X(q). \end{aligned} \tag{144}$$

The final estimate will also this time be consistent (under mild conditions) but not of optimal accuracy. Instead we get

Corollary 13. *Consider the situation in Theorem A.4 and let the model structure $\mathcal{M}$ be given by Eq. (142). Assume that $\mathcal{M}$ is identifiable at η_0 and that the data set $Z^\infty = \{y(1), r(1), \dots, y(\infty), r(\infty)\}$ is informative enough with respect to $\mathcal{M}$. Assume also that $G_0 \in \mathcal{G}$ holds for the model structure (143). Then, if the open-loop parameters θ are estimated from Eq. (144) using the Markov estimate, we have that*

$$\sqrt{N}(\hat{\rho}_N - \rho_0) \in AsN(0, P_\rho^{\mathrm{BJ}}), \tag{145a}$$

where

$$P_\rho^{\mathrm{BJ}} = \lambda_0 \left[\frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{\Phi_u^r}{|H_0|^2} G_\rho' G_\rho'^* \,\mathrm{d}\omega\right]^{-1}. \tag{145b}$$

The proof is given in Appendix B.7.

Thus

$$P_\rho^{\mathrm{BJ}} = P_\rho^{\mathrm{I}} \ (\geq P_\rho^{\mathrm{D}}), \tag{146}$$

and once again we conclude that with an independently parameterized noise model the indirect method gives worse accuracy than the direct method. The difference is quantified by the term Δ (cf. Eq. (127)) which is missing in Eq. (145).

8.3. Variance results for the projection method

In this section some results for the efficiency of the projection method will be presented. Exact analysis of the asymptotic variance of the parameter estimates is not easy in the general case and we will not go into any further details on this here. However, if we consider the model (cf. Eq. (79))

$$y(t) = G(q, \rho)\hat{u}(t) + H_*(q)e(t), \tag{147}$$

and rewrite the true system (2) as

$$y(t) = G_0(q)\hat{u}(t) + w(t), \tag{148}$$

where

$$w(t) = v(t) + G_0(q)(u(t) - \hat{u}(t)), \tag{149}$$

we can nevertheless gain some insight into the variance properties of the projection method.

With the projection method, $\hat{u}$ and w in Eq. (148) are uncorrelated. We may thus regard the second step of the projection method as an open-loop identification of G_0 using measurements of y and $\hat{u}$. The difference of the second step of the projection method and direct identification of G_0 using measurements of y and u is that

$\hat{u}$ replaces u and w replaces v. Typically, we will have

$$\Phi_{\hat{u}} < \Phi_u \quad \text{and} \quad \Phi_w > \Phi_v \tag{150}$$

implying that

$$\frac{\Phi_{\hat{u}}}{\Phi_w} < \frac{\Phi_u}{\Phi_v}. \tag{151}$$

This shows that the signal-to-noise ratio will be worse than for direct identification using y and u, and we can thus expect that the accuracy of the projection method will be suboptimal. See also Forssell and Ljung (1999, to appear).

If u is generated as in Eq. (15) and $\hat{u} = S_0 r$ (SISO) we have that $w = S_0 v$ and we are back in an open-loop situation and all standard open-loop results hold (since r and v are independent). As an example we have the following consequence of Theorem A.4 (G'_ρ is short for $\mathrm{d}/\mathrm{d}\rho G(\mathrm{e}^{\mathrm{i}\omega}, \rho)|_{\rho=\rho_0}$):

Corollary 14. *Consider the estimate $\hat{\rho}_N$ defined by Eqs.* (30)–(32) *where ε is determined from the model structure* (147) *which we assume is uniformly stable and satisfies $G_0 \in \mathscr{G}$. Suppose that Assumptions* 1 *and* 2 *hold and that the input is generated as in Eq.* (15) *and that $\hat{u} = S_0 r$. Suppose also that for a unique value ρ_0 interior to $D_{\mathscr{M}}$ we have*

$$\hat{\rho}_N \to \rho_0, \quad \text{w.p. 1 as } N \to \infty, \tag{152}$$

$$R = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{\Phi_{\hat{u}}}{|H_*|^2} G'_\rho G'^{*}_\rho \,\mathrm{d}\omega > 0, \tag{153}$$

$$\sqrt{N} E\left[\frac{1}{N}\sum_{t=1}^{N} [\psi(t,\rho_0)\varepsilon(t,\rho_0) - \bar{E}\psi(t,\rho_0)\varepsilon(t,\rho_0)]\right] \to 0, \quad \text{as } N \to \infty, \tag{154}$$

where ψ denotes the negative gradient of ε. Then

$$\sqrt{N}(\hat{\rho}_N - \rho_0) \in \mathrm{AsN}(0, P_\rho), \tag{155a}$$

$$P_\rho = R^{-1} Q R^{-1}, \tag{155b}$$

$$Q = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{\Phi_w \Phi_{\hat{u}}}{|H_*|^4} G'_\rho G'^{*}_\rho \,\mathrm{d}\omega. \tag{155c}$$

9. Summarizing discussion

We have attempted to give a status report on identification of closed-loop systems. Several methods have been studied and we have shown that most of the common methods can be viewed as special parameterizations of the general prediction error method. This allows simultaneous analysis of the statistical properties of the methods using the general prediction error theory. The main results can be summarized as follows.

The direct approach

- gives consistency and optimal accuracy, regardless of the feedback, if the noise and dynamic models contain a true description of the system.
- requires a correct noise model (possibly estimated) to avoid bias in the dynamics transfer function for closed-loop data (Corollary 5). For this reason, user chosen weightings for custom-made bias distribution cannot be used.

The indirect approach

- requires perfect knowledge of the regulator but can be applied with fixed noise models and still guarantee consistency (Corollary 6).
- is equivalent to the direct approach applied with a noise model containing the regulator (Lemma 3).

The joint input–output approach

- gives consistent estimates of the open-loop system regardless of the noise models used, as long as the feedback has a certain (linear) structure.
- can be seen as a combination of direct identification of the open-loop system and of the regulator.

The projection method

- allows the true system to be approximated in an arbitrary, user chosen frequency domain norm regardless of the feedback (Corollary 7).

Further comments:

(1) As the model orders tend to infinity, the asymptotic variance expressions are the same for all methods in case of linear feedback (Corollaries 8 and 9).
(2) For the finite model order case, the direct method meets the Cramèr–Rao bound and is thus optimal. The indirect method generally gives worse accuracy. The improved accuracy for the direct method can be traced to the fact that a constrained noise model (that includes the true noise description) allows some leverage from the noise part of the input to estimate the dynamics (Corollaries 10 and 11, and Corollaries 12 and 13).

The optimal statistical properties, the simplicity, and the general applicability of the direct method implies that this should be seen as the first choice of methods for closed-loop identification. When this method fails no other method will succeed. In the literature several other methods, e.g., the indirect method, have been advocated mainly due to their ability to shape the bias distribution by use of fixed noise models (prefilters). However, a drawback with most of these methods is that they cannot be applied to systems with arbitrary/unknown

feedback mechanisms (the indirect method, e.g., requires a perfectly known feedback law to be applicable). An exception is the projection method that works regardless of the nature of the feedback. Another drawback with the indirect and joint input–output methods is that they generally give suboptimal accuracy.

If a low order model is sought that should approximate the system dynamics in a pre-specified frequency norm there are two ways to proceed:

(1) First estimate a high-order model, with small bias, using the direct approach and then reduce this model to lower order using the proper frequency weighting.
(2) Use the projection method.

Compared to the former approach, the projection method has the advantage that a low order model can be fitted directly to the data with arbitrary frequency weighting. However, if the high-order model used in the first approach is (essentially) bias-free this will also give us explicit information about the error in the low order model, which favors this approach.

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Appendix A. Theoretical results for the prediction error method

A.1. Complement to Section 3.2

Here we will prove Eq. (39). Let us formalize the result in a theorem:

Theorem A.1. *Let $\hat{\theta}_N$ be defined by Eqs.* (30)–(32) *where $L(q, \theta) = I$. Suppose that the model structure* (27) *is uniformly stable and that Assumptions* 1 *and* 2 *hold. Then, w. p.* 1 *as $N \to \infty$,*

$$\hat{\theta}_N \to D_c = \arg\min_{\theta \in D_\mathcal{M}} \int_{-\pi}^{\pi} \left[[(G_0 - G_\theta) \quad (H_0 - H_\theta)] \Phi_{\chi_0} \times \begin{bmatrix} (G_0 - G_\theta)^* \\ (H_0 - H_\theta)^* \end{bmatrix} (H_\theta \Lambda H_\theta^*)^{-1} \right] \mathrm{d}\omega. \tag{A.1}$$

Proof. From Ljung (1978) we have that $\hat{\theta}_N$ will tend to the minimizing argument of $\bar{V}(\theta)$, w. p. 1 as $N \to \infty$. This is Eq. (37). To derive the expression (A.1) we need to derive an expression for the spectrum of the prediction error ε, cf. Eq. (38). Write H_θ as short for $H(e^{i\omega}, \theta)$, etc. Then, using Eq. (2) we get

$$\begin{aligned} \varepsilon &= H_\theta^{-1}(y - G_\theta u) \\ &= H_\theta^{-1}[(G_0 - G_\theta)u + (H_0 - H_\theta)e] + e. \end{aligned} \tag{A.2}$$

The last term is independent of the rest since $(G_0(q) - G(q, \theta))u(t)$ depends only on $e(s)$ for $s < t$ (by assumption) and $H_0(q) - H(q, \theta)$ contains a delay (since both H_0 and H are monic). Using Eq. (8) the spectrum of ε thus becomes

$$\Phi_\varepsilon = H_\theta^{-1}[(G_0 - G_\theta) \quad (H_0 - H_\theta)] \Phi_{\chi_0} \times \begin{bmatrix} (G_0 - G_\theta)^* \\ (H_0 - H_\theta)^* \end{bmatrix} H_\theta^{-*} + \Lambda_0. \tag{A.3}$$

The result now follows since the last term is independent of θ and $\mathrm{tr}[AB] = \mathrm{tr}[BA]$ for any matrices A and B of appropriate dimensions. □

A.2. Complement to Section 3.3

In this appendix we will give the exact statements of the results (43)–(45), including the necessary technical assumptions. The main results are given in Theorem A.2 and Corollary A.3 below.

First of all the vec-operator introduced in Eq. (40) is defined as ($A_{i,j}$ is the (i, j) element of the $(n \times m)$ matrix A)

$$\mathrm{vec}[A] = [A_{1,1}, A_{2,1}, \ldots, A_{n,1}, A_{1,2}, A_{2,2}, \ldots, A_{n,m}]^T. \tag{A.4}$$

The model structure considered is

$$\varepsilon(t, \theta) = y(t) - \hat{y}(t|\theta) = H^{-1}(q, \theta)(y(t) - G(q, \theta)u(t)). \tag{A.5}$$

Let

$$\theta^*(n) = \arg\min_{\theta \in D_\mathcal{M}} \bar{E} \varepsilon^T(t, \theta) \varepsilon(t, \theta). \tag{A.6}$$

(If the minimum is not unique, let $\theta^*(n)$ denote *any* minimizing element.) Now define the estimate $\hat{\theta}_N(n, \delta)$ by

$$\hat{\theta}_N(n, \delta) = \arg\min_{\theta \in D_\mathcal{M}} V_N(\theta, n, \delta, Z^N), \tag{A.7}$$

$$V_N(\theta, n, \delta, Z^N) = \frac{1}{N} \sum_{t=1}^{N} \varepsilon^T(t, \theta) \varepsilon(t, \theta) + \delta \| \theta - \theta^*(n) \|^2. \tag{A.8}$$

Here δ is a regularization parameter helping us to select a unique minimizing element in (A.7) in case $\delta = 0$ leads to nonunique minima. Let

$$T_n^*(e^{i\omega}) = T(e^{i\omega}, \theta^*(n)), \tag{A.9}$$

$$\hat{T}_N(e^{i\omega}, n, \delta) = T(e^{i\omega}, \hat{\theta}_N(n, \delta)), \tag{A.10}$$

$$T_0(e^{i\omega}) = \mathrm{vec}[G_0(e^{i\omega}) \quad H_0(e^{i\omega})]. \tag{A.11}$$

It will be assumed that

$$\lim_{n\to\infty} n^2 E\|\varepsilon(t,\theta^*(n)) - e(t)\|^2 = 0 \tag{A.12}$$

which implies that $T_n^*(e^{i\omega})$ tends to $T_0(e^{i\omega})$ as n tends to infinity. Define $Z(q,\theta)$ through (cf. Eq. (42))

$$\frac{\partial}{\partial\theta_k} T(q,\theta) = q^{-k+1}\frac{\partial}{\partial\theta_1} T(q,\theta) \triangleq q^{-k+1} Z(q,\theta). \tag{A.13}$$

Let the matrix function $Z(e^{i\omega},\theta)$ be denoted by $Z_0(e^{i\omega})$ when evaluated for $T_0(e^{i\omega})$. It will also be assumed that

$$Z_0(e^{i\omega})Z_0^T(e^{i\omega}) \tag{A.14}$$

is invertible and that

$$\lim_{N\to\infty}\frac{1}{\sqrt{N}}\sum_{t=1}^{N} E\left[\frac{d}{d\theta}\|\varepsilon(t,\theta^*(n))\|^2\right] = 0 \quad (n \text{ fixed}). \tag{A.15}$$

Finally it will be assumed that all signal spectra involved exist and are well defined and that

$$\bar{\sigma}(\Phi_u(\omega)) \le C, \quad \underline{\sigma}(\Phi_u(\omega)) \ge \lambda > 0 \quad \forall\omega. \tag{A.16}$$

The main variance result is now

Theorem A.2. *Consider the estimate $\hat{T}_N(e^{i\omega},n,\delta)$ under the assumptions* (42) *and* (A.5)–(A.16). *Suppose also that Assumptions* 1 *and* 2 *hold. Then*

$$\sqrt{N}\,\mathrm{vec}[\hat{T}_N(e^{i\omega},n,\delta) - T_n^*(e^{i\omega})] \in AsN(0,P(\omega,n,\delta)) \quad \text{as } N\to\infty \text{ for fixed } n,\delta, \tag{A.17}$$

where

$$\lim_{\delta\to 0}\lim_{n\to\infty}\frac{1}{n}P(\omega,n,\delta) = (\Phi_{\chi_0}(\omega))^{-T}\otimes\Phi_v(\omega). \tag{A.18}$$

(*Here $\otimes$ denotes the Kronecker product.*)

This was proven for the SISO case in Ljung (1985). The extension to the MIMO case was established in Zhu (1989).

The expression (43) can be seen as an intuitive, but not formally correct, interpretation of Eqs. (A.17) and (A.18).

In open loop we have (Ljung, 1985; Zhu, 1989)

Corollary A.3. *Consider the same situation as in Theorem A.2 but assume that $H(q,\theta)$ is independent of θ. Assume that $\Phi_{ue}=0$ and that Eq.* (A.12) *is relaxed to*

$$n^2\|G_n^*(e^{i\omega}) - G_0(e^{i\omega})\| \to 0 \quad \text{as } n\to\infty. \tag{A.19}$$

Then

$$\sqrt{N}\,\mathrm{vec}[\hat{G}_N(e^{i\omega},n,\delta) - G_n^*(e^{i\omega})] \in AsN(0,P(\omega,n,\delta)) \quad \text{as } N\to\infty \text{ for fixed } n,\delta, \tag{A.20}$$

where

$$\lim_{\delta\to 0}\lim_{n\to\infty}\frac{1}{n}P(\omega,n,\delta) = (\Phi_u(\omega))^{-T}\otimes\Phi_v(\omega). \tag{A.21}$$

A.3. Complement to Section 3.4

We will now turn to the question of asymptotic distribution of the parameter vector estimates. For notational convenience the discussion will be limited to the SISO case. Let λ_0 denote the variance of the noise e in Eq. (2) in the SISO case. The following result is a variant of Theorem 9.1 in Ljung (1987):

Theorem A.4. *Consider the estimate $\hat{\theta}_N$ determined by Eqs.* (30)–(32) *with $L(q,\theta)=1$. Assume that the model structure is linear and uniformly stable and that Assumptions* 1 *and* 2 *hold. Suppose that for a unique value θ^* interior to $D_{\mathcal{M}}$ we have*

$$\hat{\theta}_N \to \theta^*, \quad w.p.\ 1 \text{ as } N\to\infty, \tag{A.22}$$

$$R = \bar{V}''(\theta^*) > 0, \tag{A.23}$$

$$\sqrt{N}E\left[\frac{1}{N}\sum_{t=1}^{N}[\psi(t,\theta^*)\varepsilon(t,\theta^*) - \bar{E}\psi(t,\theta^*)\varepsilon(t,\theta^*)]\right]\to 0, \quad \text{as } N\to\infty. \tag{A.24}$$

Then

$$\sqrt{N}(\hat{\theta}_N - \theta^*) \in AsN(0,P_\theta), \tag{A.25a}$$

$$P_\theta = R^{-1}QR^{-1}, \tag{A.25b}$$

$$Q = \lim_{N\to\infty} N\cdot E\{[V_N'(\theta^*,Z^N)][V_N'(\theta^*,Z^N)]^T\}. \tag{A.25c}$$

Remark. Frequency domain expressions for R and Q can easily be derived using Parseval's relationship (cf. Ljung, 1987, Section 9.4).

Appendix B. Additional proofs

B.1. Proof of Corollary 5

The result is direct consequence of Theorem A.1. To derive the expression (101) we simply have to insert the factorization (10) of Φ_{χ_0} into Eq. (A.1).

We now turn to the proof of Eq. (103) Consider the first term in Eq. (101). If we define $\tilde{G}_\theta = G_0 - G_\theta$ we have (neglecting the trace operator and the scaling Λ^{-1} which are irrelevant here)

$$\int_{-\pi}^{\pi} H_\theta^{-1}(\tilde{G}_\theta + B_\theta)\Phi_u(\tilde{G}_\theta + B_\theta)^* H_\theta^{-*}\,d\omega = \int_{-\pi}^{\pi}(H_\theta^{-1}\tilde{G}_\theta M_u + H_\theta^{-1}B_\theta M_u)$$

$$\times (H_\theta^{-1}\tilde{G}_\theta M_u + H_\theta^{-1}B_\theta M_u)^*\,\mathrm{d}\omega$$
$$= \int_{-\pi}^{\pi} (H_\theta^{-1}\tilde{G}_\theta M_u + [H_\theta^{-1}B_\theta M_u]_+)$$
$$\times (H_\theta^{-1}\tilde{G}_\theta M_u + [H_\theta^{-1}B_\theta M_u]_+)^*$$
$$+ [H_\theta^{-1}B_\theta M_u]_- [H_\theta^{-1}B_\theta M_u]_-^*\,\mathrm{d}\omega$$
$$= \int_{-\pi}^{\pi} H_\theta^{-1}\{(\tilde{G}_\theta + B_\theta^+)\Phi_u(\tilde{G}_\theta + B_\theta^+)^*$$
$$+ B_\theta^-\Phi_u(B_\theta^-)^*\}H_\theta^{-*}\,\mathrm{d}\omega, \tag{B.1}$$

where we in the second step used that $H_\theta^{-1}\tilde{G}_\theta M_u$ is causal. Eq. (103) follows.

B.2. Proof of Corollary 6

The main convergence result was given in Theorem A.1. To derive the expression (109) we need to compute Φ_ε. The prediction error is (using Eq. (25))

$$\varepsilon = H_*^{-1}(y - G_\theta^c r)$$
$$= H_*^{-1}[(G_0^c - G_\theta^c)r + S_0 v + G_0^c d]. \tag{B.2}$$

Thus

$$\Phi_\varepsilon = H_*^{-1}[(G_0^c - G_\theta^c)\Phi_r(G_0^c - G_\theta^c)^* + (G_0^c - G_\theta^c)\Phi_{rd}(G_0^c)^*$$
$$+ G_0^c\Phi_{dr}(G_0^c - G_\theta^c)^*]H_*^{-*} + \theta\text{-independent terms.} \tag{B.3}$$

With $\bar{B}$ given by Eq. (110) this can be rewritten as

$$\Phi_\varepsilon = H_*^{-1}[(G_0^c + \bar{B} - G_\theta^c)\Phi_r(G_0^c + \bar{B} - G_\theta^c)^*]H_*^{-*}$$
$$+ \theta\text{-independent terms.} \tag{B.4}$$

Eq. (109) follows if we consider only the θ-dependent terms.

The alternative expression (111) will now be derived. Define $\tilde{G}_\theta^c = G_0^c - G_\theta^c$. Then

$$\int_{-\pi}^{\pi} H_*^{-1}(\tilde{G}_\theta^c + \bar{B})\Phi_r(\tilde{G}_\theta^c + \bar{B})^* H_*^{-*}\,\mathrm{d}\omega$$
$$= \int_{-\pi}^{\pi} (H_*^{-1}\tilde{G}_\theta^c M_r + H_*^{-1}\bar{B}M_r)$$
$$\times (H_*^{-1}\tilde{G}_\theta^c M_r + H_*^{-1}\bar{B}M_r)^*\,\mathrm{d}\omega$$
$$= \int_{-\pi}^{\pi} (H_*^{-1}\tilde{G}_\theta^c M_r + [H_*^{-1}\bar{B}M_r]_+)$$
$$\times (H_*^{-1}\tilde{G}_\theta M_r + [H_*^{-1}\bar{B}M_r]_+)^*$$
$$+ [H_*^{-1}\bar{B}M_r]_- [H_*^{-1}\bar{B}M_r]_-^*\,\mathrm{d}\omega$$
$$= \int_{-\pi}^{\pi} H_*^{-1}(\tilde{G}_\theta^c + \bar{B}^+)\Phi_r(\tilde{G}_\theta^c + \bar{B}^+)^* H_*^{-*}\,\mathrm{d}\omega + C, \tag{B.5}$$

where $\bar{B}^+$ is given by Eq. (112) and C is a constant matrix, independent of θ. In the second step used that $H_*^{-1}\tilde{G}_\theta^c M_r$ is causal. It follows that Eq. (111) is equivalent to Eq. (109), as claimed.

B.3. Proof of Corollary 7

The proof of Eq. (116) is analogous to the corresponding proof for the indirect approach and hence omitted. For the proof of Eq. (119) we first note that the main convergence result follows from Theorem A.1. We need to compute Φ_ε. The prediction error is

$$\varepsilon = H_*^{-1}(y - G_\theta\hat{u}) = H_*^{-1}[(G_0 - G_\theta)\hat{u} + w(t)], \tag{B.6}$$

where $w = G_0\tilde{u} + v$. Hence

$$\Phi_\varepsilon = H_*^{-1}[(G_0 - G_\theta)\Phi_{\hat{u}}(G_0 - G_\theta)^* + (G_0 - G_\theta)\Phi_{w\hat{u}}$$
$$+ \Phi_{\hat{u}w}(G_0 - G_\theta)^*]H_*^{-*} + (H_*)^{-1}\Phi_w H_*^{-*}, \tag{B.7}$$

where we can note that $\Phi_{w\hat{u}} = G_0\Phi_{\tilde{u}\hat{u}}$. If we use $\tilde{B}$ defined in Eq. (120) we can thus alternatively write Φ_ε as

$$\Phi_\varepsilon = H_*^{-1}[(G_0 + \tilde{B} - G_\theta)\Phi_{\hat{u}}(G_0 + \tilde{B} - G_\theta)^*]H_*^{-*}$$
$$+ \theta\text{-independent terms} \tag{B.8}$$

from which Eq. (119) follows.

B.4. Proof of Corollary 9

For the proof of the result for the indirect approach we will assume that the model is given by Eq. (49). (The parameterization as well as the noise model can be arbitrary. Thus the result applies to all variants of the indirect method studied in this paper.) According to Corollary A.3 the random variable

$$\sqrt{N}\,\mathrm{vec}[\hat{G}_N^c(\mathrm{e}^{\mathrm{i}\omega}, n, \delta) - G_n^{c,*}(\mathrm{e}^{\mathrm{i}\omega})] \tag{B.9}$$

will have an asymptotic normal distribution with covariance matrix $\bar{P}(\omega, n, \delta)$ satisfying

$$\lim_{\delta\to 0}\lim_{n\to\infty}\frac{1}{n}\bar{P}(\omega, n, \delta) = (\Phi_r(\omega))^{-\mathrm{T}} \otimes \Phi_{v_c}(\omega), \tag{B.10}$$

where $\Phi_{v_c} = S_0\Phi_v S_0^*$. From the Taylor expansion

$$\mathrm{vec}[\hat{G}_N(\mathrm{e}^{\mathrm{i}\omega}, n, \delta) - G_n^*(\mathrm{e}^{\mathrm{i}\omega})]$$
$$= \left(\frac{\mathrm{d}\,\mathrm{vec}[G]}{\mathrm{d}\,\mathrm{vec}[G^c]}(\mathrm{e}^{\mathrm{i}\omega}, \theta^*(n))\right)^{\mathrm{T}}$$
$$\times \{\mathrm{vec}[\hat{G}_N^c(\mathrm{e}^{\mathrm{i}\omega}, n, \delta) - G_n^{c,*}(\mathrm{e}^{\mathrm{i}\omega})]\}$$
$$+ O(\|\mathrm{vec}[\hat{G}_N^c(\mathrm{e}^{\mathrm{i}\omega}, n, \delta) - G_n^{c,*}(\mathrm{e}^{\mathrm{i}\omega})]\|), \tag{B.11}$$

we see that the covariance matrix of

$$\sqrt{N}\,\mathrm{vec}[\hat{G}_N(\mathrm{e}^{\mathrm{i}\omega}, n, \delta) - G_n^*(\mathrm{e}^{\mathrm{i}\omega})] \tag{B.12}$$

will be

$$P(\omega, n, \delta) = \left(\frac{\mathrm{d}\,\mathrm{vec}[G]}{\mathrm{d}\,\mathrm{vec}[G^{\mathrm{c}}]}\,(\mathrm{e}^{\mathrm{i}\omega}, \theta^*(n))\right)^{-\mathrm{T}} \bar{P}(\omega, n, \delta) \times \left(\frac{\mathrm{d}\,\mathrm{vec}[G]}{\mathrm{d}\,\mathrm{vec}[G^{\mathrm{c}}]}\,(\mathrm{e}^{-\mathrm{i}\omega}, \theta^*(n))\right)^{-1}. \tag{B.13}$$

Note that

$$G^{\mathrm{c}} = SG = (I + GK)^{-1}G = G - GKG + GKGKG - \cdots. \tag{B.14}$$

Repeated use of the formula (Graham, 1981)

$$\frac{\mathrm{d}\,\mathrm{vec}[AXB]}{\mathrm{d}\,\mathrm{vec}[X]} = (B \otimes A^{\mathrm{T}}) \tag{B.15}$$

and the chain rule now gives

$$\frac{\mathrm{d}\,\mathrm{vec}[G^{\mathrm{c}}]}{\mathrm{d}\,\mathrm{vec}[G]} = I - (KG \otimes I) - (I \otimes K^{\mathrm{T}}G^{\mathrm{T}}) + (KG \otimes K^{\mathrm{T}}G^{\mathrm{T}}) + (KGKG \otimes I) + (I \otimes K^{\mathrm{T}}G^{\mathrm{T}}K^{\mathrm{T}}G^{\mathrm{T}}) - \cdots. \tag{B.16}$$

As can be readily verified, with $S^{\mathrm{i}} = (I + KG)^{-1}$ and $S = (I + GK)^{-1}$, we thus have that

$$\frac{\mathrm{d}\,\mathrm{vec}[G^{\mathrm{c}}]}{\mathrm{d}\,\mathrm{vec}[G]} = (S^{\mathrm{i}} \otimes S^{\mathrm{T}}) = ((S^{\mathrm{i}})^{\mathrm{T}} \otimes S)^{\mathrm{T}}. \tag{B.17}$$

Consequently

$$\begin{aligned}\lim_{n\to\infty} \frac{1}{n} P(\omega, n, \delta) &= ((S_0^{\mathrm{i}})^{\mathrm{T}} \otimes S_0)^{-1}((\Phi_r)^{-\mathrm{T}} \otimes \Phi_{v_{\mathrm{c}}})((S_0^{\mathrm{i}})^{\mathrm{T}} \otimes S_0)^{-*} \\ &= ((S_0^{\mathrm{i}})^{-\mathrm{T}} \otimes S_0^{-1})((\Phi_r)^{-\mathrm{T}} \otimes \Phi_{v_{\mathrm{c}}})(((S_0^{\mathrm{i}})^{\mathrm{T}})^{-*} \otimes S_0^{-*}) \\ &= ((S_0^{\mathrm{i}})^{-\mathrm{T}}(\Phi_r)^{-\mathrm{T}}((S_0^{\mathrm{i}})^*)^{-\mathrm{T}}) \otimes (S_0^{-1}\Phi_{v_{\mathrm{c}}}S_0^{-*}) \\ &= (\Phi_u^r)^{-\mathrm{T}} \otimes \Phi_v \end{aligned} \tag{B.18}$$

which concludes the proof for the indirect approach.

For the joint input–output approach we will in this proof assume that the model is

$$\begin{bmatrix} \hat{y}(t, \theta) \\ \hat{u}(t, \theta) \end{bmatrix} = \begin{bmatrix} G^{\mathrm{c}}(q, \theta) \\ S^{\mathrm{i}}(q, \theta) \end{bmatrix} r(t) \tag{B.19}$$

and that the open-loop estimate is computed as $\hat{G}_N = \hat{G}_N^{\mathrm{c}}(\hat{S}_N^{\mathrm{i}})^{-1}$. (The result holds regardless of the parameterization and regardless of the noise model used, hence it holds for all joint input–output method studied in this paper, including the two-stage/projection method.) Write Q as short for $[(G^{\mathrm{c}})^{\mathrm{T}}\ (S^{\mathrm{i}})^{\mathrm{T}}]^{\mathrm{T}}$. According to Corollary A.3 the random variable

$$\sqrt{N}\,\mathrm{vec}[\hat{Q}_N(\mathrm{e}^{\mathrm{i}\omega}, n, \delta) - Q_n^*(\mathrm{e}^{\mathrm{i}\omega})] \tag{B.20}$$

will have an asymptotic normal distribution with covariance matrix $\bar{P}(\omega, n, \delta)$ satisfying

$$\lim_{\delta\to 0}\lim_{n\to\infty} \frac{1}{n}\bar{P}(\omega, n, \delta) = (\Phi_r(\omega))^{-\mathrm{T}} \otimes \begin{bmatrix} I \\ -K(\mathrm{e}^{\mathrm{i}\omega}) \end{bmatrix} \Phi_{v_{\mathrm{c}}}(\omega) \times \begin{bmatrix} I \\ -K(\mathrm{e}^{\mathrm{i}\omega}) \end{bmatrix}^* \tag{B.21}$$

where $\Phi_{v_{\mathrm{c}}} = S_0\Phi_v S_0^*$. From the Taylor expansion

$$\begin{aligned}\mathrm{vec}[\hat{G}_N(\mathrm{e}^{\mathrm{i}\omega}, n, \delta) - G_n^*(\mathrm{e}^{\mathrm{i}\omega})] &= \left(\frac{\mathrm{d}\,\mathrm{vec}[G]}{\mathrm{d}\,\mathrm{vec}[Q]}\,(\mathrm{e}^{\mathrm{i}\omega}, \theta^*(n))\right)^{\mathrm{T}} \\ &\quad \times \{\mathrm{vec}[\hat{Q}_N(\mathrm{e}^{\mathrm{i}\omega}, n, \delta) - Q_n^*(\mathrm{e}^{\mathrm{i}\omega})]\} \\ &\quad + O(\|\,\mathrm{vec}[\hat{Q}_N(\mathrm{e}^{\mathrm{i}\omega}, n, \delta) - Q_n^*(\mathrm{e}^{\mathrm{i}\omega})]\,\|),\end{aligned} \tag{B.22}$$

we see that the covariance matrix of

$$\sqrt{N}\,\mathrm{vec}[\hat{G}_N(\mathrm{e}^{\mathrm{i}\omega}, n, \delta) - G_n^*(\mathrm{e}^{\mathrm{i}\omega})] \tag{B.23}$$

will be

$$P(\omega, n, \delta) = \left(\frac{\mathrm{d}\,\mathrm{vec}[G]}{\mathrm{d}\,\mathrm{vec}[Q]}\,(\mathrm{e}^{\mathrm{i}\omega}, \theta^*(n))\right)^{-\mathrm{T}} \bar{P}(\omega, n, \delta) \times \left(\frac{\mathrm{d}\,\mathrm{vec}[G]}{\mathrm{d}\,\mathrm{vec}[Q]}\,(\mathrm{e}^{-\mathrm{i}\omega}, \theta^*(n))\right)^{-1} \tag{B.24}$$

By use of the formula (B.15) and the chain rule it can be shown that

$$\begin{aligned}\frac{\mathrm{d}\,\mathrm{vec}[G]}{\mathrm{d}\,\mathrm{vec}[Q]} &= ((S^{\mathrm{i}})^{-1} \otimes [I \quad 0]^{\mathrm{T}}) + ((S^{\mathrm{i}})^{-1} \otimes [0 \quad -G^{\mathrm{c}}(S^{\mathrm{i}})^{-1}]^{\mathrm{T}}) \\ &= (S^{\mathrm{i}})^{-1} \otimes [I \quad -G^{\mathrm{c}}(S^{\mathrm{i}})^{-1}]^{\mathrm{T}}.\end{aligned} \tag{B.25}$$

Note that $G^{\mathrm{c}}(S^{\mathrm{i}})^{-1} \to G_0$ as $n \to \infty$. Thus

$$\begin{aligned}\lim_{n\to\infty} \frac{1}{n} P(\omega, n, \delta) &= ((S_0^{\mathrm{i}})^{-\mathrm{T}} \otimes [I \quad -G_0]) \times \left((\Phi_r)^{-\mathrm{T}} \otimes \begin{bmatrix} I \\ -K \end{bmatrix} \Phi_{v_{\mathrm{c}}} \begin{bmatrix} I \\ -K \end{bmatrix}^*\right) \\ &\quad \times ((S_0^{\mathrm{i}})^{-\mathrm{T}} \otimes [I \quad -G_0])^* \\ &= ((S_0^{\mathrm{i}})^{-\mathrm{T}}(\Phi_r)^{-\mathrm{T}}((S_0^{\mathrm{i}})^*)^{-\mathrm{T}}) \\ &\quad \otimes (S_0^{-1}\Phi_{v_{\mathrm{c}}}S_0^{-*}) = (\Phi_u^r)^{-\mathrm{T}} \otimes \Phi_v\end{aligned} \tag{B.26}$$

which concludes the proof for the joint input–output approach and the also proof of Corollary 9.

B.5. Proof of Corollary 10

Under the conditions in the corollary we have that $\hat{\theta}_N \to \theta_0$ w. p. 1 as $N \to \infty$, see, e.g., Theorem 8.3 in Ljung (1987). It follows from Theorem A.4 that

$$\sqrt{N}(\hat{\theta}_N - \theta_0) \in AsN(0, P_\theta), \tag{B.27}$$

where

$$P_\theta = \lambda_0 R_\theta, \tag{B.28}$$

$$R_\theta = [\bar{E}\psi(t,\theta_0)\psi^{\mathrm{T}}(t,\theta_0)]. \tag{B.29}$$

Using Parseval's relationship, R_θ can be written as (see, e.g., Ljung, 1987)

$$R_\theta = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{1}{|H_0|^2}\,\bar{T}'_\theta \Phi_{\chi_0} \bar{T}'^{*}_\theta \,\mathrm{d}\omega, \tag{B.30}$$

where $\bar{T} = [G\ \ H]$ and where $\bar{T}'_\theta = (\mathrm{d}/\mathrm{d}\theta)\bar{T}|_{\theta=\theta_0}$. From Eq. (13) it follows that

$$\bar{T}'_\theta \Phi_{\chi_0} \bar{T}'^{*}_\theta = \bar{T}'_\theta \Phi^r_{\chi_0} \bar{T}'^{*}_\theta + \bar{T}'_\theta \Phi^e_{\chi_0} \bar{T}'^{*}_\theta. \tag{B.31}$$

We may thus write

$$R_\theta = R^r_\theta + R^e_\theta, \tag{B.32}$$

where, due to the chosen parameterization,

$$R^r_\theta = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{1}{|H_0|^2}\,\bar{T}'_\theta \Phi^r_{\chi_0} \bar{T}'^{*}_\theta \,\mathrm{d}\omega = \begin{bmatrix} R^r_\rho & 0 \\ 0 & 0 \end{bmatrix} \tag{B.33}$$

and

$$R^e_\theta = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{1}{|H_0|^2}\,\bar{T}'_\theta \Phi^e_{\chi_0} \bar{T}'^{*}_\theta \,\mathrm{d}\omega = \begin{bmatrix} R^e_\rho & R^e_{\rho\beta} \\ R^e_{\beta\rho} & R^e_\beta \end{bmatrix}. \tag{B.34}$$

Straight-forward calculations will also show that

$$R^r_\rho = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{\Phi^r_u}{|H_0|^2}\,G'_\rho G'^{*}_\rho \,\mathrm{d}\omega \tag{B.35}$$

and

$$\begin{aligned} R^e_\theta &= \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{1}{|H_0|^2}\,\bar{T}'_\theta \Phi^e_{\chi_0} \bar{T}'^{*}_\theta \,\mathrm{d}\omega \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{\lambda_0}{|H_0|^2} \begin{bmatrix} KS_0H_0G'_\rho \\ -H'_\beta \end{bmatrix} \begin{bmatrix} KS_0H_0G'_\rho \\ -H'_\beta \end{bmatrix}^{*} \mathrm{d}\omega. \end{aligned} \tag{B.36}$$

P_ρ, the covariance matrix for $\hat{\rho}_N$ can be found as the top left block of P_θ. Combining Eqs. (B.28), (B.29), (B.33), and (B.34) we thus have that

$$P_\rho = \lambda_0 (R^r_\rho + \Delta)^{-1}, \tag{B.37}$$

$$\Delta = R^e_\rho - R^e_{\rho\beta}(R^e_\beta)^{-1} R^e_{\beta\rho}, \tag{B.38}$$

where R^r_ρ is given by Eq. (B.35) and expressions for R^e_ρ, $R^e_{\rho\beta}$, and R^e_β can be extracted from Eq. (B.36).

To prove the lower bound in Eq. (128) note that Δ is the Schur complement of R^e_β in the matrix R^e_θ given by Eq. (B.34). Since $R^e_\theta \geq 0$ we thus have $\Delta \geq 0$. Next, define

$$\psi_e(t,\theta_0) = \bar{T}'_\theta(q,\theta_0) \begin{bmatrix} K(q)S_0(q)H_0(q) \\ -1 \end{bmatrix} e(t). \tag{B.39}$$

Then we can alternatively write R^e_θ as

$$R^e_\theta = \bar{E}\psi_e(t,\theta_0)\psi_e^{\mathrm{T}}(t,\theta_0). \tag{B.40}$$

Now introduce the notation

$$w(t) = H_0^{-1}(q)e(t), \tag{B.41}$$

$$\bar{G}'_\rho(q) = K(q)S_0(q)H_0(q)G'_\rho(q), \tag{B.42}$$

so that

$$\psi_e(t,\theta_0) = \begin{bmatrix} \bar{G}'_\rho(q)w(t) \\ -H'_\beta(q)w(t) \end{bmatrix}. \tag{B.43}$$

Here the number of rows in $\bar{G}'_\rho$ and H'_β are consistent with the partitioning (B.33), (B.34). From the well-known least squares projections (see, e.g., Anderson and Moore, 1979), we recognize Δ as *the error covariance matrix when estimating* $\bar{G}'_\rho w$ *from* $H'_\beta w$. Suppose now that the order of the noise model tends to infinity. Without loss of generality we can assume that

$$H(q,\beta) = 1 + \lim_{M\to\infty} \sum_{k=1}^{M} h_k q^{-k}. \tag{B.44}$$

Then knowing $H'_\beta w$ is equivalent to knowing all past w. Then $\bar{G}'_\rho w$ can be determined exactly from $H'_\beta w$, and $\Delta = 0$. At the other extreme, a fixed (and correct) noise model will make $\Delta = R^e_\rho = \bar{E}\bar{G}'_\rho w(\bar{G}'_\rho w)^{\mathrm{T}}$, which is the largest value Δ may have. This proves the upper bound in Eq. (128) and the remaining statements in the corollary.

B.6. Proof of Corollary 12

Let

$$\begin{aligned} A(q) &= 1 + a_1 q^{-1} + \cdots + a_{n_a} q^{-n_a}, \\ B(q) &= b_1 q^{-1} + \cdots + b_{n_b} q^{-n_b}, \\ C(q) &= 1 + c_1 q^{-1} + \cdots + c_{n_c} q^{-n_c}, \end{aligned} \tag{B.45}$$

and similarly for the closed-loop polynomials. Define θ as

$$\theta = [a_1, \ldots, a_{n_a}, b_1, \ldots, b_{n_b}, c_1, \ldots, c_{n_c}]^{\mathrm{T}} \tag{B.46}$$

and η equivalently. Furthermore, let the regulator polynomials be

$$\begin{aligned} X(q) &= x_0 + x_1 q^{-1} + \cdots + x_{n_x} q^{-n_x}, \\ Y(q) &= 1 + y_1 q^{-1} + \cdots + y_{n_y} q^{-n_y}. \end{aligned} \tag{B.47}$$

With these definitions the system of equations (140) can be written as

$$\Gamma\theta = \bar{\eta}, \tag{B.48}$$

where

$$\Gamma = \begin{bmatrix} \Gamma_Y & \Gamma_X & 0 \\ 0 & \Gamma_Y & 0 \\ 0 & 0 & \Gamma_Y \end{bmatrix}, \qquad \Gamma_Y = \begin{bmatrix} 1 & & \\ y_1 & \ddots & \\ \vdots & \ddots & 1 \\ & & y_1 \\ & & \vdots \end{bmatrix},$$

$$\Gamma_X = \begin{bmatrix} x_0 & & \\ x_1 & \ddots & \\ \vdots & \ddots & x_0 \\ & & x_1 \\ & & \vdots \end{bmatrix}, \tag{B.49}$$

while

$$\bar{\eta} = [a_{c,1} - y_1, a_{c,2} - y_2, \dots, b_{c,1}, b_{c,2}, \dots, c_{c,1} - y_1, c_{c,2} - y_2, \dots]^{\mathrm{T}}. \tag{B.50}$$

Now, if the closed-loop parameters η are estimated from a set of N data we have from Theorem A.4 that

$$\sqrt{N}(\hat{\eta}_N - \eta_0) \in AsN(0, P_\eta), \tag{B.51}$$

$$P_\eta = \lambda_0[\bar{E}\psi_{\mathrm{c}}(t, \eta_0)\psi_{\mathrm{c}}^{\mathrm{T}}(t, \eta_0)]^{-1}, \tag{B.52}$$

$$\psi_{\mathrm{c}}(t, \eta) = -\frac{\mathrm{d}}{\mathrm{d}\eta}\frac{A^{\mathrm{c}}(q)}{C^{\mathrm{c}}(q)}\left(y(t) - \frac{B^{\mathrm{c}}(q)}{A^{\mathrm{c}}(q)}r(t)\right). \tag{B.53}$$

Consider Eq. (B.48). The Markov estimate of θ is

$$\hat{\theta}_N = [\Gamma^{\mathrm{T}}(\mathrm{Cov}\,\hat{\bar{\eta}}_N)^{-1}\Gamma]^{-1}\Gamma^{\mathrm{T}}(\mathrm{Cov}\,\hat{\bar{\eta}}_N)^{-1}\hat{\bar{\eta}}_N \tag{B.54}$$

from which it follows that

$$\sqrt{N}(\hat{\theta}_N - \theta_0) \in AsN(0, P_\theta), \tag{B.55}$$

$$P_\theta = N[\Gamma^{\mathrm{T}}(\mathrm{Cov}\,\hat{\bar{\eta}}_N)^{-1}\Gamma]^{-1}. \tag{B.56}$$

We also have that

$$\mathrm{Cov}\,\hat{\bar{\eta}}_N = \frac{1}{N}P_\eta = \frac{\lambda_0}{N}[\bar{E}\psi_{\mathrm{c}}(t, \eta_0)\psi_{\mathrm{c}}^{\mathrm{T}}(t, \eta_0)]^{-1}. \tag{B.57}$$

Hence

$$P_\theta = \lambda_0[\bar{E}\Gamma^{\mathrm{T}}\psi_{\mathrm{c}}(t, \eta_0)\psi_{\mathrm{c}}^{\mathrm{T}}(t, \eta_0)\Gamma]^{-1}. \tag{B.58}$$

The gradient vector $\psi_{\mathrm{c}}(t, \eta_0)$ is

$$\psi_{\mathrm{c}}(t, \eta_0) = \frac{1}{C_0^{\mathrm{c}}(q)}[-y(t-1), \dots, -y(t-n_{a_{\mathrm{c}}}), r(t-1), \dots, r(t-n_{b_{\mathrm{c}}}), e(t-1), \dots, e(t-n_{c_{\mathrm{c}}})]^{\mathrm{T}} \tag{B.59}$$

and since

$$\frac{Y(q)}{C_0^{\mathrm{c}}(q)} = \frac{1}{C_0(q)}$$

and

$$\frac{X(q)y(t)}{C_0^{\mathrm{c}}(q)} - \frac{Y(q)r(t)}{C_0^{\mathrm{c}}(q)} = -\frac{1}{C_0(q)}u(t), \tag{B.60}$$

we have

$$\Gamma^{\mathrm{T}}\psi_{\mathrm{c}}(t, \eta_0) = \frac{1}{C_0(q)}[-y(t-1), \dots, -y(t-n_a), u(t-1), \dots, u(t-n_b), e(t-1), \dots, e(t-n_{\mathrm{c}})]^{\mathrm{T}}. \tag{B.61}$$

It follows that

$$\Gamma^{\mathrm{T}}\psi_{\mathrm{c}}(t, \eta_0) = -\frac{\mathrm{d}}{\mathrm{d}\theta}\frac{A(q)}{C(q)}\left(y(t) - \frac{B(q)}{A(q)}u(t)\right), \tag{B.62}$$

where the right-hand side should be evaluated at $\theta = \theta_0$. But the expression on the right is equal to the negative gradient of the open-loop prediction error, that is $\psi(t, \theta)$, evaluated at $\theta = \theta_0$. Thus $\Gamma^{\mathrm{T}}\psi_{\mathrm{c}}(t, \eta_0) = \psi(t, \theta_0)$ and

$$P_\theta = \lambda_0[\bar{E}\psi(t, \theta_0)\psi^{\mathrm{T}}(t, \theta_0)]^{-1} \tag{B.63}$$

which together with Eq. (B.55) proves the corollary.

B.7. Proof of Corollary 13

Let

$$\begin{aligned} B(q) &= b_1 q^{-1} + \cdots + b_{n_b}q^{-n_b}, \\ F(q) &= 1 + f_1 q^{-1} + \cdots + f_{n_f}q^{-n_f}, \\ C(q) &= 1 + c_1 q^{-1} + \cdots + c_{n_{\mathrm{c}}}q^{-n_{\mathrm{c}}}, \\ D(q) &= 1 + d_1 q^{-1} + \cdots + d_{n_d}q^{-n_d}, \end{aligned} \tag{B.64}$$

and similarly for the closed-loop polynomials and let X, Y, Γ_X, and Γ_Y be as in the proof of Corollary 12. Then Eq. (144) can be written

$$\tilde{\Gamma}\rho = \bar{\rho}_\eta, \tag{B.65}$$

where

$$\tilde{\Gamma} = \begin{bmatrix} \Gamma_Y & 0 \\ \Gamma_X & \Gamma_Y \end{bmatrix},$$

$$\bar{\rho}_\eta = [b_{\mathrm{c},1}, b_{\mathrm{c},2}, \dots, f_{\mathrm{c},1} - y_1, f_{\mathrm{c},2} - y_2, \dots]^{\mathrm{T}}. \tag{B.66}$$

Using the same arguments as in the proof of Corollary 12 we get

$$\sqrt{N}(\hat{\rho}_N - \rho_0) \in AsN(0, P_\rho) \tag{B.67}$$

$$P_\rho = N[\tilde{\Gamma}^{\mathrm{T}}(\mathrm{Cov}\,\hat{\bar{\rho}}_{\eta,N})^{-1}\tilde{\Gamma}]^{-1}. \tag{B.68}$$

We also have that

$$\operatorname{Cov}\hat{\bar{\rho}}_{\eta,N} = \operatorname{Cov}\hat{\rho}_{\eta,N} = \frac{\lambda_0}{N}\left[\bar{E}\psi_{c,\rho_\eta}(t,\eta_0)\psi_{c,\rho_\eta}^{T}(t,\eta_0)\right]^{-1}, \tag{B.69}$$

where ψ_{c,ρ_η} is the negative gradient of the closed-loop predictions error,

$$\varepsilon_c(t,\eta) = \frac{1}{H^c(q,\beta_\eta)}\left(y(t) - G^c(q,\rho_\eta)r(t)\right) \tag{B.70}$$

$$= \frac{1}{H^c(q,\beta_\eta)}\left(y(t) - \frac{B^c(q)}{F^c(q)}r(t)\right) \tag{B.71}$$

taken w.r.t. ρ_η, giving

$$\psi_{c,\rho_\eta}(t,\eta_0) = \frac{1}{H_0^c(q)F_0^c(q)}\left[r(t-1),\ \dots,r(t-n_{b_c}),\ -\frac{B_0^c(q)}{F_0^c(q)}\left(r(t-1),\ \dots,r(t-n_{f_c})\right)\right]. \tag{B.72}$$

Returning to P_ρ we thus have that

$$P_\rho = \lambda_0\left[\bar{E}\tilde{\Gamma}^{T}\psi_c(t,\eta_0)\psi_c^{T}(t,\eta_0)\tilde{\Gamma}\right]^{-1}. \tag{B.73}$$

However, since

$$H_0^c = S_0H_0, \qquad \frac{F_0^cY - B_0^cX}{(F_0^c)^2} = \frac{S_0^2}{F_0}, \qquad \frac{B_0^cY}{(F_0^c)^2} = \frac{S_0^2}{F_0}\cdot\frac{B_0}{F_0} \tag{B.74}$$

we get (after some calculations)

$$\tilde{\Gamma}^{T}\psi_{c,\rho_\eta}(t,\eta_0) = \frac{S_0(q)}{H_0(q)F_0(q)}\left[r(t-1),\ \dots,r(t-n_b),\ -\frac{B_0(q)}{F_0(q)}\left(r(t-1),\ \dots,r(t-n_f)\right)\right]^{T}. \tag{B.75}$$

So

$$\tilde{\Gamma}^{T}\psi_{c,\rho_\eta}(t,\eta_0) = -\frac{S_0(q)}{H_0(q)}\frac{d}{d\rho_\theta}\left(y(t) - \frac{B(q)}{F(q)}r(t)\right) \tag{B.76}$$

$$= \frac{S_0(q)}{H_0(q)}\frac{d}{d\rho_\theta}G(q,\rho_\theta)r(t), \tag{B.77}$$

where the derivatives on the right should be evaluated at $\rho_\theta = \rho_{\theta,0}$. Using Parseval's formula we thus have that

$$P_\rho = \lambda_0\left[\frac{1}{2\pi}\int_{-\pi}^{\pi}\frac{\Phi_u^r}{|H_0|^2}G_\rho' G_\rho'^{*}\,d\omega\right]^{-1} \tag{B.78}$$

and the corollary follows.

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